

# Analysis

March 12, 2025

## 1 Analysis

### 1.1 Setup

```
[24]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import scienceplots
import scipy.stats as stats
import seaborn as sns
from scikit_posthocs import posthoc_dunn

plt.style.use(['science', 'ieee'])
# plt.style.use(['science'])

[25]: arb_df = pd.read_csv('../arb-contracts-final-results.csv')
eth_df = pd.read_csv('../eth-contracts-final-results.csv')
ftm_df = pd.read_csv('../ftm-contracts-final-results.csv')
opt_df = pd.read_csv('../opt-contracts-final-results.csv')
pol_df = pd.read_csv('../pol-contracts-final-results.csv')

# add column to each df to indicate the blockchain
arb_df['Blockchain'] = 'arb'
eth_df['Blockchain'] = 'eth'
ftm_df['Blockchain'] = 'ftm'
opt_df['Blockchain'] = 'opt'
pol_df['Blockchain'] = 'pol'

blockchains = ['arb', 'eth', 'ftm', 'opt', 'pol']
combined_df = pd.concat([arb_df, eth_df, ftm_df, opt_df, pol_df])
combined_df.to_csv('../combined-results.csv')

[26]: def create_sub_df(df, keyword):
    new_df = df.filter(regex=f'^({keyword}|Blockchain)')
    new_df.columns = new_df.columns.str.replace(keyword, '')
    return new_df

combined_metrics_df = create_sub_df(combined_df, 'metrics.')
```

```
combined_patterns_df = create_sub_df(combined_df, 'patterns.')

# rename Storage Optimization column to Storage Saver
combined_patterns_df = combined_patterns_df.rename(columns={'Storage_
↳Optimization': 'Storage Saver'})
```

```
[27]: def get_stats(df):
    df_stats = df.describe()
    df_stats.loc['mode'] = df.mode().iloc[0]
    df_stats = df_stats.rename(index={'50%': 'median'})
    df_stats = df_stats.drop('count')
    return df_stats

combined_stats = get_stats(combined_metrics_df)
```

```
[28]: print('#####')
print('overall')
print(combined_stats)

for blockchain, group in combined_metrics_df.groupby('Blockchain'):
    print('#####')
    print(blockchain)
    print(get_stats(group))
```

```
#####
```

```
overall
```

|        | Contract Lines of Code | External Function Count | Inheritance Depth \ |
|--------|------------------------|-------------------------|---------------------|
| mean   | 106.482882             | 4.910463                | 0.851916            |
| std    | 212.832993             | 10.960443               | 1.417819            |
| min    | 1.000000               | 0.000000                | 0.000000            |
| 25%    | 17.000000              | 0.000000                | 0.000000            |
| median | 53.000000              | 1.000000                | 0.000000            |
| 75%    | 123.000000             | 6.000000                | 1.000000            |
| max    | 13933.000000           | 470.000000              | 11.000000           |
| mode   | 12.000000              | 0.000000                | 0.000000            |

|        | Internal Function Count | Max Local Variables \ |
|--------|-------------------------|-----------------------|
| mean   | 6.843356                | 4.205814              |
| std    | 17.300010               | 3.127734              |
| min    | 0.000000                | 0.000000              |
| 25%    | 0.000000                | 2.000000              |
| median | 2.000000                | 4.000000              |
| 75%    | 8.000000                | 6.000000              |
| max    | 1184.000000             | 108.000000            |
| mode   | 0.000000                | 4.000000              |

|      | Mean Cyclomatic Complexity | Mean Local Variables | Number of Functions \ |
|------|----------------------------|----------------------|-----------------------|
| mean | 0.674645                   | 2.189069             | 14.586150             |

|        |           |           |             |
|--------|-----------|-----------|-------------|
| std    | 0.737741  | 1.415455  | 24.720232   |
| min    | 0.000000  | 0.000000  | 0.000000    |
| 25%    | 0.000000  | 1.000000  | 3.000000    |
| median | 0.820000  | 2.050000  | 7.000000    |
| 75%    | 1.040000  | 3.000000  | 15.000000   |
| max    | 29.000000 | 38.000000 | 1184.000000 |
| mode   | 0.000000  | 1.000000  | 1.000000    |

|        | Private Function Count | Public Function Count | State Variable Count \ |
|--------|------------------------|-----------------------|------------------------|
| mean   | 0.536200               | 2.296130              | 1.866199               |
| std    | 1.619795               | 5.583016              | 5.764575               |
| min    | 0.000000               | 0.000000              | 0.000000               |
| 25%    | 0.000000               | 0.000000              | 0.000000               |
| median | 0.000000               | 0.000000              | 0.000000               |
| 75%    | 0.000000               | 1.000000              | 2.000000               |
| max    | 32.000000              | 106.000000            | 271.000000             |
| mode   | 0.000000               | 0.000000              | 0.000000               |

|        | Total Cyclomatic Complexity |
|--------|-----------------------------|
| mean   | 13.241612                   |
| std    | 29.956682                   |
| min    | 0.000000                    |
| 25%    | 0.000000                    |
| median | 3.000000                    |
| 75%    | 15.000000                   |
| max    | 2892.000000                 |
| mode   | 0.000000                    |

#####

arb

|        | Contract Lines of Code | External Function Count | Inheritance Depth \ |
|--------|------------------------|-------------------------|---------------------|
| mean   | 95.089264              | 5.601378                | 1.192218            |
| std    | 175.124638             | 9.504110                | 1.780319            |
| min    | 1.000000               | 0.000000                | 0.000000            |
| 25%    | 18.000000              | 0.000000                | 0.000000            |
| median | 50.000000              | 2.000000                | 0.000000            |
| 75%    | 108.000000             | 8.000000                | 2.000000            |
| max    | 10997.000000           | 373.000000              | 10.000000           |
| mode   | 17.000000              | 0.000000                | 0.000000            |

|        | Internal Function Count | Max Local Variables \ |
|--------|-------------------------|-----------------------|
| mean   | 7.765720                | 4.484162              |
| std    | 18.727472               | 2.945376              |
| min    | 0.000000                | 0.000000              |
| 25%    | 0.000000                | 2.000000              |
| median | 3.000000                | 4.000000              |
| 75%    | 9.000000                | 6.000000              |
| max    | 1184.000000             | 108.000000            |
| mode   | 0.000000                | 4.000000              |

|        | Mean Cyclomatic Complexity | Mean Local Variables | Number of Functions \ |
|--------|----------------------------|----------------------|-----------------------|
| mean   | 0.651906                   | 2.270774             | 16.564378             |
| std    | 0.731563                   | 1.362342             | 26.457697             |
| min    | 0.000000                   | 0.000000             | 0.000000              |
| 25%    | 0.000000                   | 1.330000             | 3.000000              |
| median | 0.790000                   | 2.160000             | 8.000000              |
| 75%    | 1.000000                   | 3.000000             | 19.000000             |
| max    | 27.000000                  | 17.000000            | 1184.000000           |
| mode   | 0.000000                   | 2.000000             | 1.000000              |

|        | Private Function Count | Public Function Count | State Variable Count \ |
|--------|------------------------|-----------------------|------------------------|
| mean   | 0.408319               | 2.788961              | 2.264045               |
| std    | 1.328851               | 6.112084              | 6.907590               |
| min    | 0.000000               | 0.000000              | 0.000000               |
| 25%    | 0.000000               | 0.000000              | 0.000000               |
| median | 0.000000               | 0.000000              | 0.000000               |
| 75%    | 0.000000               | 3.000000              | 2.000000               |
| max    | 27.000000              | 85.000000             | 269.000000             |
| mode   | 0.000000               | 0.000000              | 0.000000               |

|        | Total Cyclomatic Complexity |
|--------|-----------------------------|
| mean   | 14.211865                   |
| std    | 35.668146                   |
| min    | 0.000000                    |
| 25%    | 0.000000                    |
| median | 4.000000                    |
| 75%    | 16.000000                   |
| max    | 2892.000000                 |
| mode   | 0.000000                    |

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eth

|        | Contract Lines of Code | External Function Count | Inheritance Depth \ |
|--------|------------------------|-------------------------|---------------------|
| mean   | 100.696024             | 4.817489                | 0.632393            |
| std    | 191.788504             | 9.443872                | 1.127022            |
| min    | 1.000000               | 0.000000                | 0.000000            |
| 25%    | 17.000000              | 0.000000                | 0.000000            |
| median | 44.000000              | 1.000000                | 0.000000            |
| 75%    | 111.000000             | 6.000000                | 1.000000            |
| max    | 13244.000000           | 371.000000              | 11.000000           |
| mode   | 5.000000               | 0.000000                | 0.000000            |

|        | Internal Function Count | Max Local Variables \ |
|--------|-------------------------|-----------------------|
| mean   | 5.434175                | 4.241696              |
| std    | 12.948462               | 3.172024              |
| min    | 0.000000                | 0.000000              |
| 25%    | 0.000000                | 2.000000              |
| median | 2.000000                | 4.000000              |

|      |            |           |
|------|------------|-----------|
| 75%  | 7.000000   | 6.000000  |
| max  | 385.000000 | 39.000000 |
| mode | 0.000000   | 4.000000  |

|        | Mean Cyclomatic Complexity | Mean Local Variables | Number of Functions \ |
|--------|----------------------------|----------------------|-----------------------|
| mean   | 0.687110                   | 2.219033             | 12.960027             |
| std    | 0.722708                   | 1.406319             | 20.184514             |
| min    | 0.000000                   | 0.000000             | 0.000000              |
| 25%    | 0.000000                   | 1.000000             | 3.000000              |
| median | 0.890000                   | 2.100000             | 6.000000              |
| 75%    | 1.080000                   | 3.000000             | 13.000000             |
| max    | 13.000000                  | 14.000000            | 385.000000            |
| mode   | 0.000000                   | 1.000000             | 1.000000              |

|        | Private Function Count | Public Function Count | State Variable Count \ |
|--------|------------------------|-----------------------|------------------------|
| mean   | 0.603584               | 2.104779              | 2.143517               |
| std    | 1.856593               | 5.140855              | 7.137326               |
| min    | 0.000000               | 0.000000              | 0.000000               |
| 25%    | 0.000000               | 0.000000              | 0.000000               |
| median | 0.000000               | 0.000000              | 0.000000               |
| 75%    | 0.000000               | 1.000000              | 1.000000               |
| max    | 27.000000              | 106.000000            | 271.000000             |
| mode   | 0.000000               | 0.000000              | 0.000000               |

|        | Total Cyclomatic Complexity |
|--------|-----------------------------|
| mean   | 11.734234                   |
| std    | 23.321817                   |
| min    | 0.000000                    |
| 25%    | 0.000000                    |
| median | 3.000000                    |
| 75%    | 13.000000                   |
| max    | 384.000000                  |
| mode   | 0.000000                    |

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ftm

|        | Contract Lines of Code | External Function Count | Inheritance Depth \ |
|--------|------------------------|-------------------------|---------------------|
| mean   | 129.012569             | 4.448368                | 0.565325            |
| std    | 220.750711             | 9.373081                | 1.019450            |
| min    | 1.000000               | 0.000000                | 0.000000            |
| 25%    | 17.000000              | 0.000000                | 0.000000            |
| median | 67.000000              | 1.000000                | 0.000000            |
| 75%    | 152.000000             | 6.000000                | 1.000000            |
| max    | 13933.000000           | 470.000000              | 9.000000            |
| mode   | 10.000000              | 0.000000                | 0.000000            |

|      | Internal Function Count | Max Local Variables \ |
|------|-------------------------|-----------------------|
| mean | 6.559865                | 4.106899              |
| std  | 15.057863               | 3.338562              |

|        |            |           |
|--------|------------|-----------|
| min    | 0.000000   | 0.000000  |
| 25%    | 0.000000   | 2.000000  |
| median | 2.000000   | 4.000000  |
| 75%    | 8.000000   | 5.000000  |
| max    | 385.000000 | 54.000000 |
| mode   | 0.000000   | 4.000000  |

|        | Mean Cyclomatic Complexity | Mean Local Variables | Number of Functions \ |
|--------|----------------------------|----------------------|-----------------------|
| mean   | 0.643924                   | 2.156687             | 13.238643             |
| std    | 0.723701                   | 1.421859             | 20.113817             |
| min    | 0.000000                   | 0.000000             | 0.000000              |
| 25%    | 0.000000                   | 1.000000             | 3.000000              |
| median | 0.730000                   | 2.000000             | 8.000000              |
| 75%    | 1.040000                   | 3.000000             | 17.000000             |
| max    | 29.000000                  | 38.000000            | 470.000000            |
| mode   | 0.000000                   | 1.000000             | 6.000000              |

|        | Private Function Count | Public Function Count | State Variable Count \ |
|--------|------------------------|-----------------------|------------------------|
| mean   | 0.528597               | 1.701813              | 1.322769               |
| std    | 1.725519               | 4.613617              | 3.692247               |
| min    | 0.000000               | 0.000000              | 0.000000               |
| 25%    | 0.000000               | 0.000000              | 0.000000               |
| median | 0.000000               | 0.000000              | 0.000000               |
| 75%    | 0.000000               | 0.000000              | 1.000000               |
| max    | 27.000000              | 71.000000             | 132.000000             |
| mode   | 0.000000               | 0.000000              | 0.000000               |

|        | Total Cyclomatic Complexity |
|--------|-----------------------------|
| mean   | 12.485621                   |
| std    | 23.920171                   |
| min    | 0.000000                    |
| 25%    | 0.000000                    |
| median | 2.000000                    |
| 75%    | 15.000000                   |
| max    | 384.000000                  |
| mode   | 0.000000                    |

#####

opt

|        | Contract Lines of Code | External Function Count | Inheritance Depth \ |
|--------|------------------------|-------------------------|---------------------|
| mean   | 110.796886             | 4.749340                | 0.650580            |
| std    | 328.033573             | 18.086118               | 1.136504            |
| min    | 1.000000               | 0.000000                | 0.000000            |
| 25%    | 18.000000              | 0.000000                | 0.000000            |
| median | 55.000000              | 1.000000                | 0.000000            |
| 75%    | 126.000000             | 5.000000                | 1.000000            |
| max    | 13933.000000           | 460.000000              | 8.000000            |
| mode   | 21.000000              | 0.000000                | 0.000000            |

|        | Internal Function Count | Max Local Variables \ |
|--------|-------------------------|-----------------------|
| mean   | 6.802502                | 3.953564              |
| std    | 20.559428               | 3.126769              |
| min    | 0.000000                | 0.000000              |
| 25%    | 0.000000                | 1.000000              |
| median | 2.000000                | 4.000000              |
| 75%    | 8.000000                | 5.000000              |
| max    | 385.000000              | 65.000000             |
| mode   | 0.000000                | 1.000000              |

|        | Mean Cyclomatic Complexity | Mean Local Variables | Number of Functions \ |
|--------|----------------------------|----------------------|-----------------------|
| mean   | 0.714576                   | 2.126846             | 13.793659             |
| std    | 0.817814                   | 1.508970             | 29.766062             |
| min    | 0.000000                   | 0.000000             | 0.000000              |
| 25%    | 0.000000                   | 1.000000             | 2.000000              |
| median | 0.940000                   | 2.000000             | 7.000000              |
| 75%    | 1.150000                   | 3.000000             | 13.000000             |
| max    | 27.000000                  | 18.000000            | 460.000000            |
| mode   | 0.000000                   | 1.000000             | 1.000000              |

|        | Private Function Count | Public Function Count | State Variable Count \ |
|--------|------------------------|-----------------------|------------------------|
| mean   | 0.640459               | 1.601358              | 1.538640               |
| std    | 1.699429               | 4.775666              | 4.346175               |
| min    | 0.000000               | 0.000000              | 0.000000               |
| 25%    | 0.000000               | 0.000000              | 0.000000               |
| median | 0.000000               | 0.000000              | 0.000000               |
| 75%    | 0.000000               | 1.000000              | 1.000000               |
| max    | 32.000000              | 80.000000             | 168.000000             |
| mode   | 0.000000               | 0.000000              | 0.000000               |

|        | Total Cyclomatic Complexity |
|--------|-----------------------------|
| mean   | 12.528792                   |
| std    | 28.296246                   |
| min    | 0.000000                    |
| 25%    | 0.000000                    |
| median | 3.000000                    |
| 75%    | 15.000000                   |
| max    | 384.000000                  |
| mode   | 0.000000                    |

#####

pol

|        | Contract Lines of Code | External Function Count | Inheritance Depth \ |
|--------|------------------------|-------------------------|---------------------|
| mean   | 107.048053             | 4.185107                | 0.788435            |
| std    | 175.914326             | 9.167650                | 1.186480            |
| min    | 1.000000               | 0.000000                | 0.000000            |
| 25%    | 17.000000              | 0.000000                | 0.000000            |
| median | 56.000000              | 1.000000                | 0.000000            |
| 75%    | 119.000000             | 6.000000                | 1.000000            |

|      |             |            |          |
|------|-------------|------------|----------|
| max  | 2986.000000 | 373.000000 | 9.000000 |
| mode | 9.000000    | 0.000000   | 0.000000 |

|        |                         |                       |
|--------|-------------------------|-----------------------|
|        | Internal Function Count | Max Local Variables \ |
| mean   | 6.415460                | 3.901338              |
| std    | 16.770051               | 3.171772              |
| min    | 0.000000                | 0.000000              |
| 25%    | 0.000000                | 2.000000              |
| median | 3.000000                | 4.000000              |
| 75%    | 8.000000                | 5.000000              |
| max    | 384.000000              | 38.000000             |
| mode   | 0.000000                | 4.000000              |

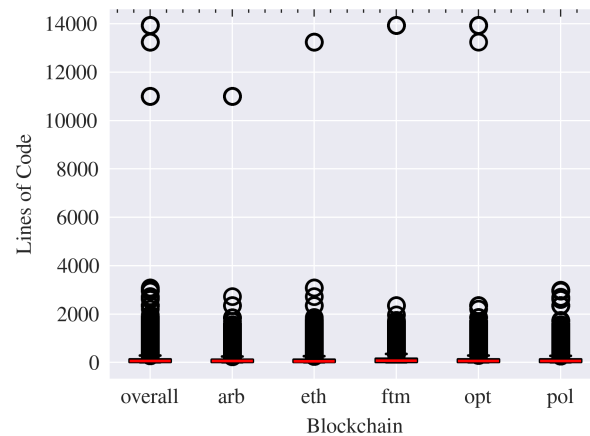
|        |                            |                      |                       |
|--------|----------------------------|----------------------|-----------------------|
|        | Mean Cyclomatic Complexity | Mean Local Variables | Number of Functions \ |
| mean   | 0.714413                   | 2.078881             | 13.872402             |
| std    | 0.710171                   | 1.439465             | 24.391023             |
| min    | 0.000000                   | 0.000000             | 0.000000              |
| 25%    | 0.000000                   | 1.000000             | 2.000000              |
| median | 0.920000                   | 2.000000             | 7.000000              |
| 75%    | 1.110000                   | 2.710000             | 14.000000             |
| max    | 11.000000                  | 14.000000            | 385.000000            |
| mode   | 0.000000                   | 1.000000             | 1.000000              |

|        |                        |                       |                        |
|--------|------------------------|-----------------------|------------------------|
|        | Private Function Count | Public Function Count | State Variable Count \ |
| mean   | 0.674623               | 2.597212              | 1.657262               |
| std    | 1.755464               | 6.148184              | 4.506337               |
| min    | 0.000000               | 0.000000              | 0.000000               |
| 25%    | 0.000000               | 0.000000              | 0.000000               |
| median | 0.000000               | 0.000000              | 0.000000               |
| 75%    | 0.000000               | 1.000000              | 1.000000               |
| max    | 27.000000              | 106.000000            | 168.000000             |
| mode   | 0.000000               | 0.000000              | 0.000000               |

|        |                             |
|--------|-----------------------------|
|        | Total Cyclomatic Complexity |
| mean   | 13.804699                   |
| std    | 28.786064                   |
| min    | 0.000000                    |
| 25%    | 0.000000                    |
| median | 4.000000                    |
| 75%    | 15.000000                   |
| max    | 384.000000                  |
| mode   | 0.000000                    |

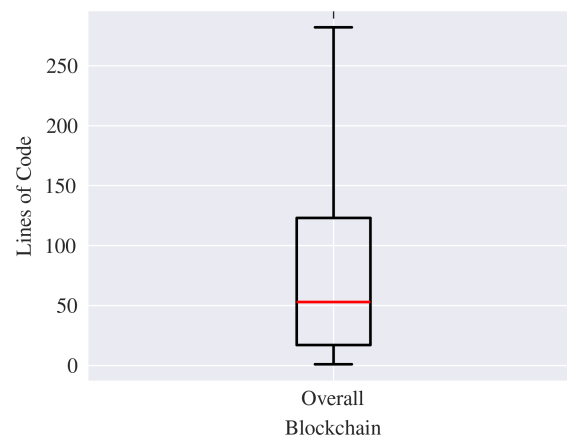
## 1.2 Metric Boxplots

```
[29]: fig = plt.figure()
plt.boxplot(combined_metrics_df['Contract Lines of Code'], showfliers=True,
            tick_labels=['Overall'])
plt.ylabel('Lines of Code')
plt.xlabel('Blockchain')
plt.show()
```



```
[30]: fig = plt.figure()
# plt.title(title)

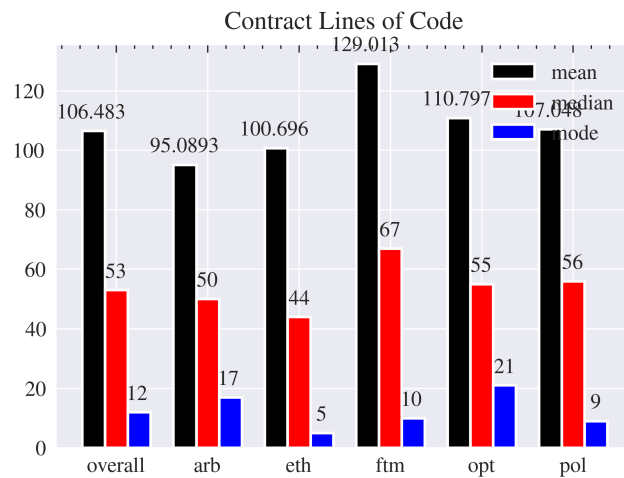
plt.boxplot(combined_metrics_df['Contract Lines of Code'], showfliers=False,
            tick_labels=['Overall'])
plt.ylabel('Lines of Code')
plt.xlabel('Blockchain')
plt.show()
```

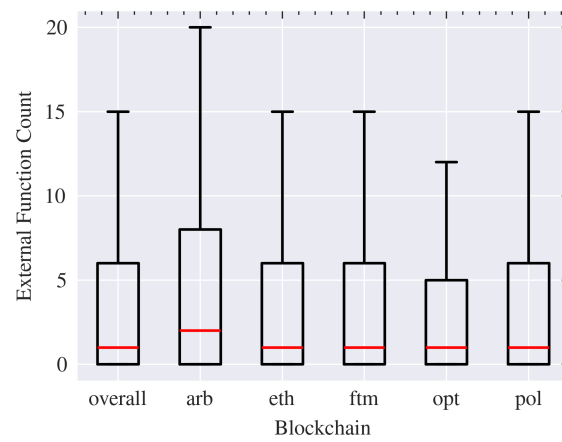
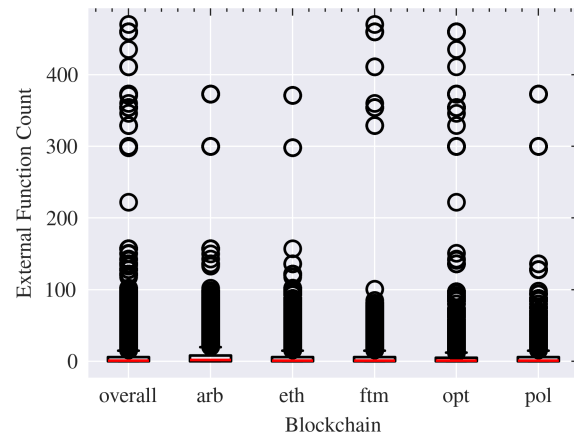
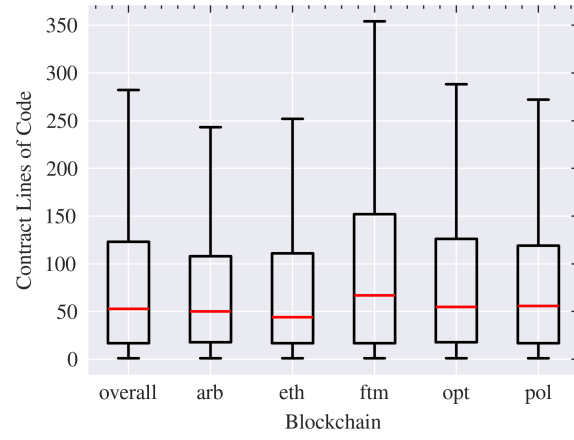


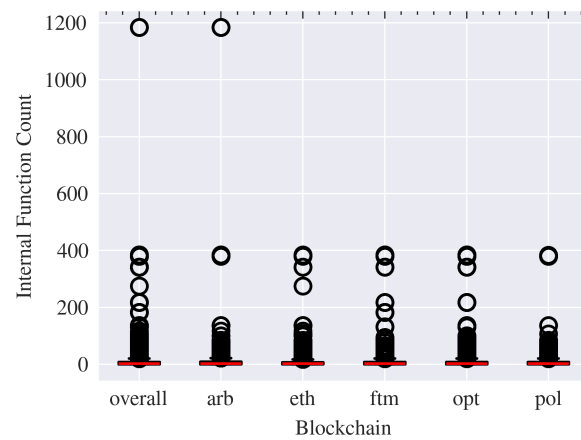
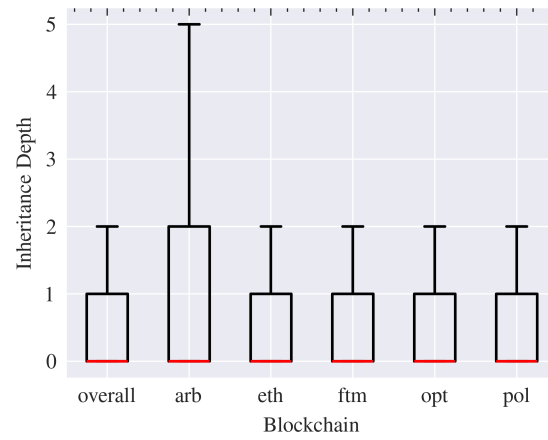
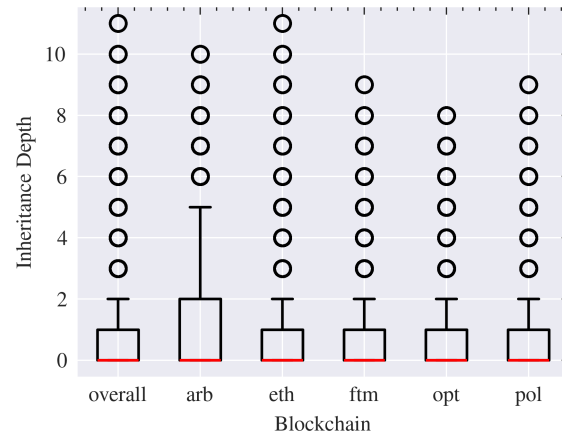
```
[31]: def create_boxplot(df, column, title, show_outliers=True):
    fig = plt.figure()
    # plt.title(title)

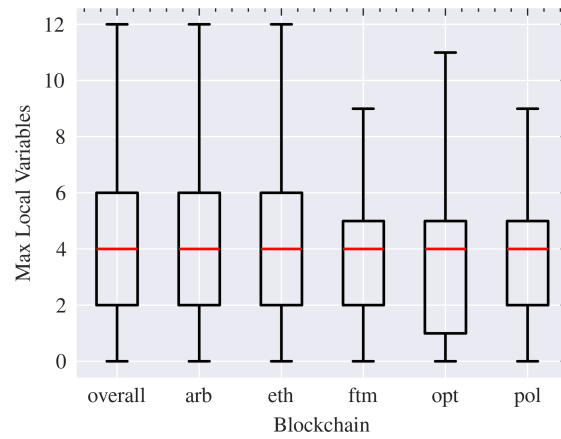
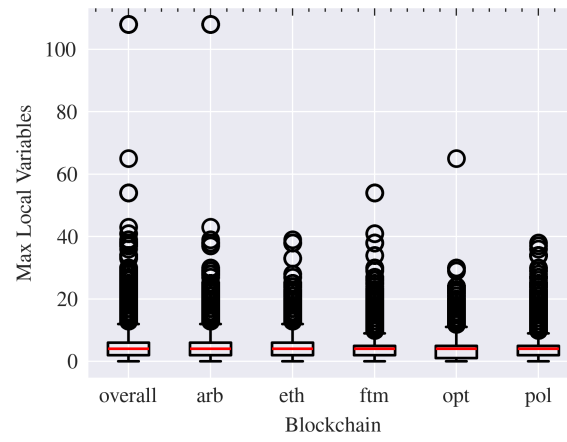
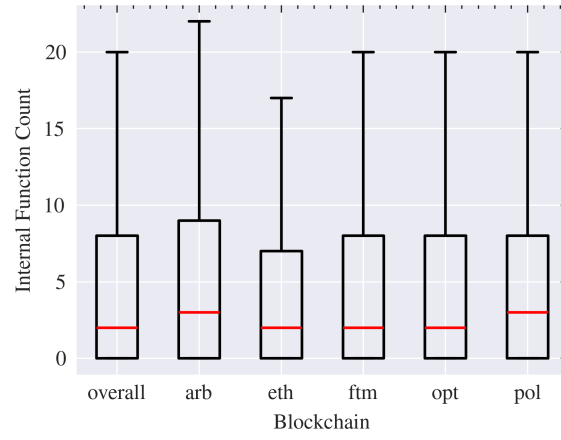
    data = [df[column]]
    labels = ['overall']
    for name, group in df.groupby('Blockchain'):
        data.append(group[column])
        labels.append(name)
    plt.boxplot(data, tick_labels=labels, showfliers=show_outliers)
    plt.ylabel(metric)
    plt.xlabel('Blockchain')
    plt.show()

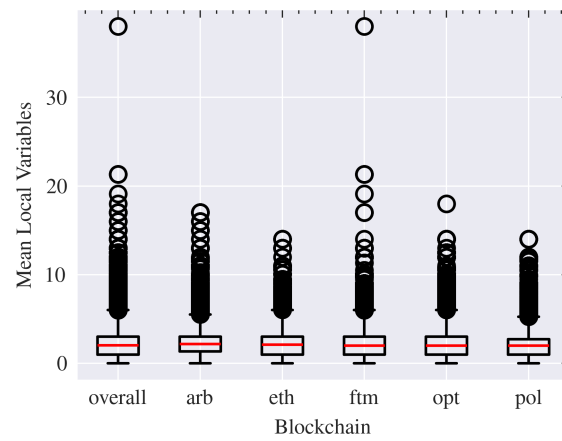
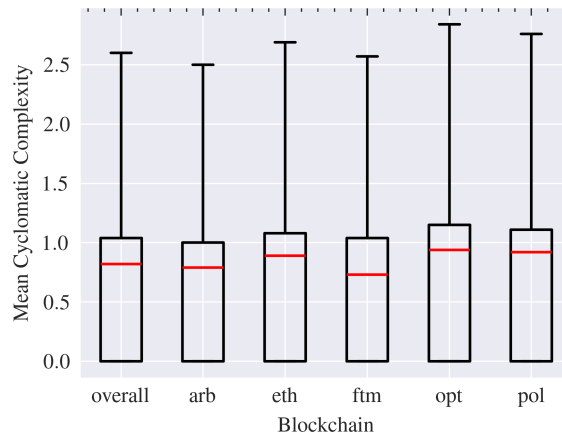
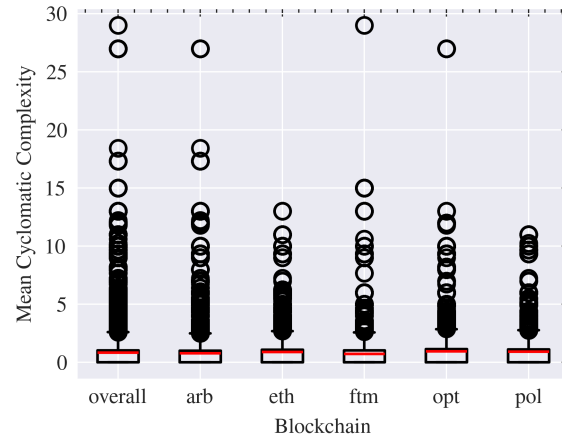
for metric in combined_stats.columns:
    create_boxplot(combined_metrics_df, metric, metric)
    create_boxplot(combined_metrics_df, metric, metric + ' - Outliers Removed',
                    False)
```

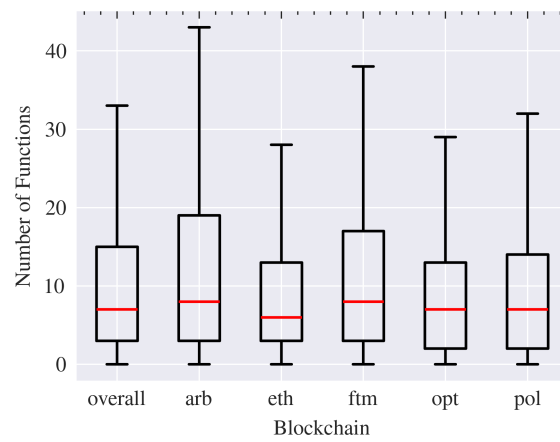
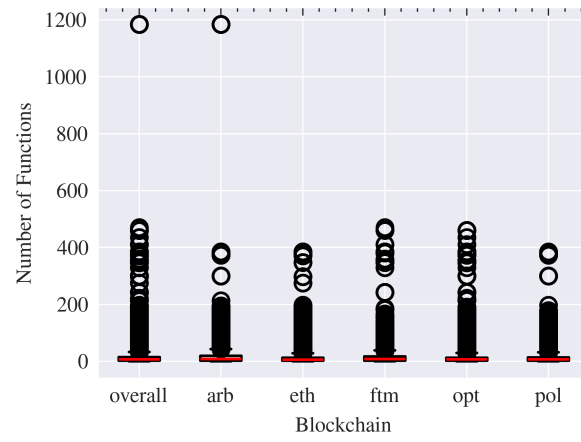
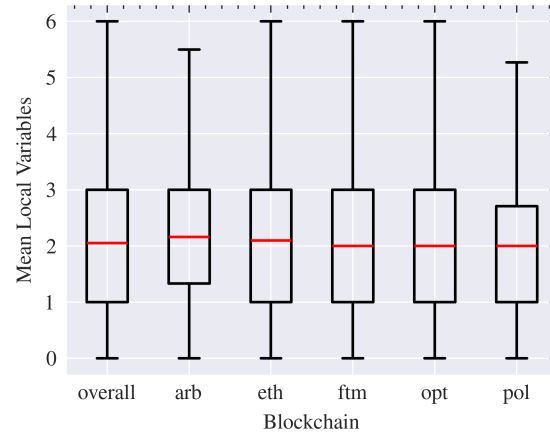


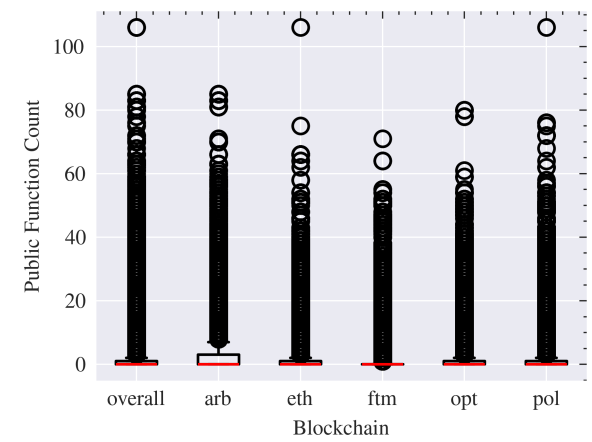
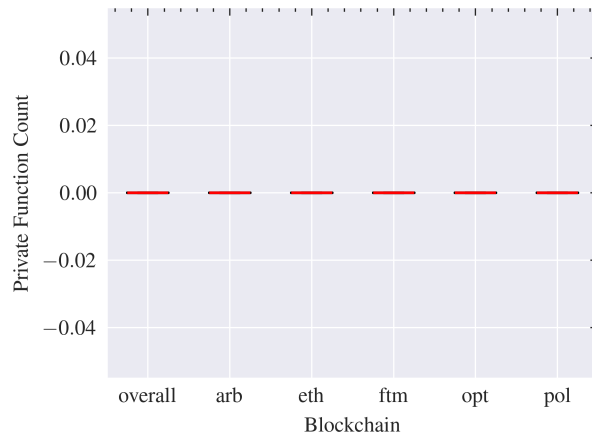
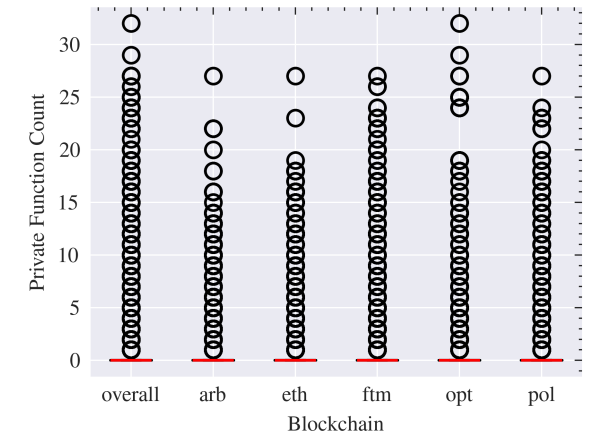


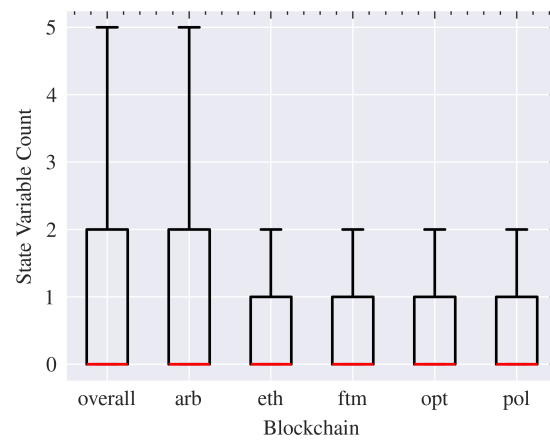
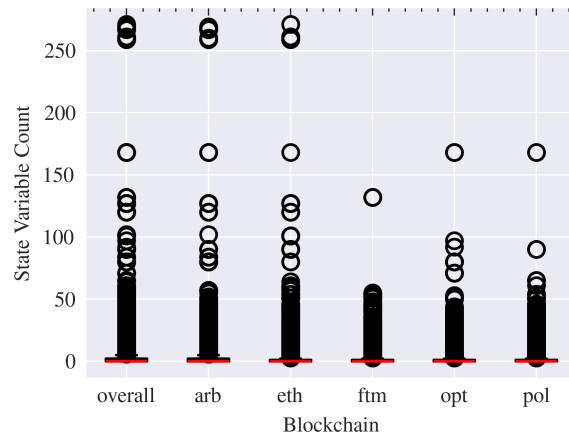
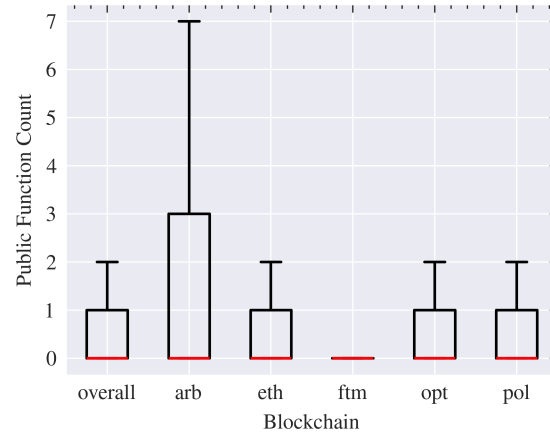


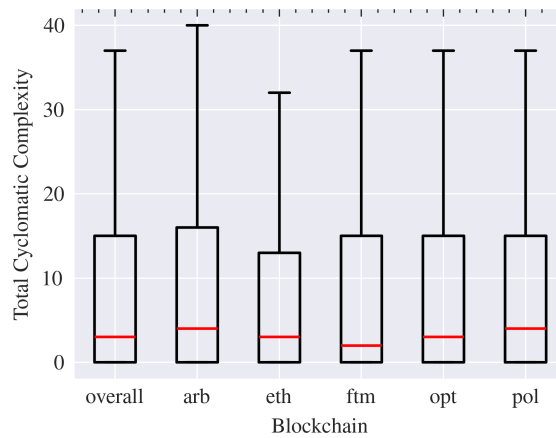
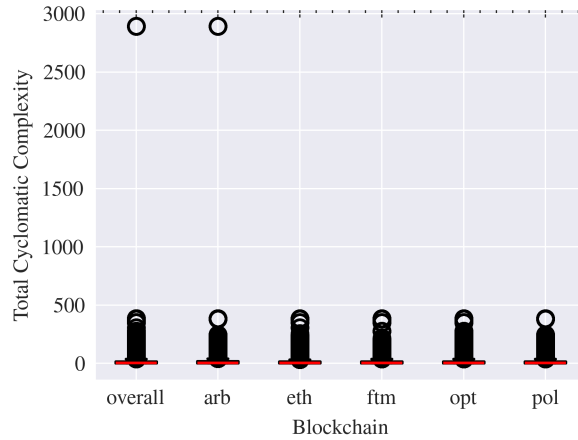












### 1.3 Average Metrics

```
[32]: def plot_averages(df, metric, title):
    means = [df[metric].mean()]
    medians = [df[metric].median()]
    modes = [df[metric].mode().iloc[0]]
    labels = ['overall']

    for name, group in df.groupby('Blockchain'):
        means.append(group[metric].mean())
        medians.append(group[metric].median())
        modes.append(group[metric].mode().iloc[0])
        labels.append(name)

    data = {
        'mean': means,
```

```

        'median': medians,
        'mode': modes
    }

    label_loc = np.arange(len(labels))
    width = 0.25 # the width of the bars
    multiplier = 0

    fig, ax = plt.subplots(layout='constrained')

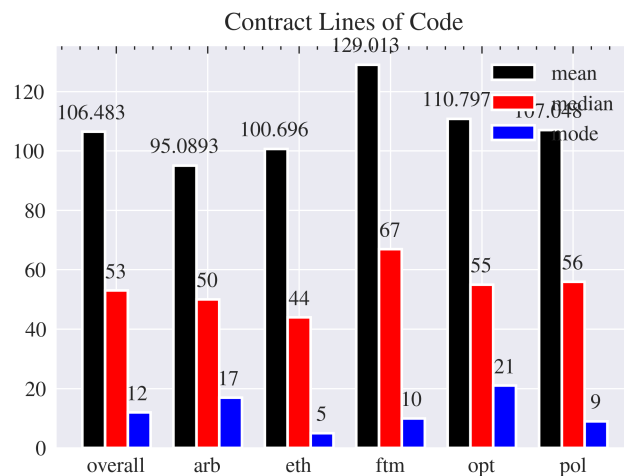
    for key, value in data.items():
        offset = width * multiplier
        rects = ax.bar(label_loc + offset, value, width, label=key)
        ax.bar_label(rects, padding=3)
        multiplier += 1

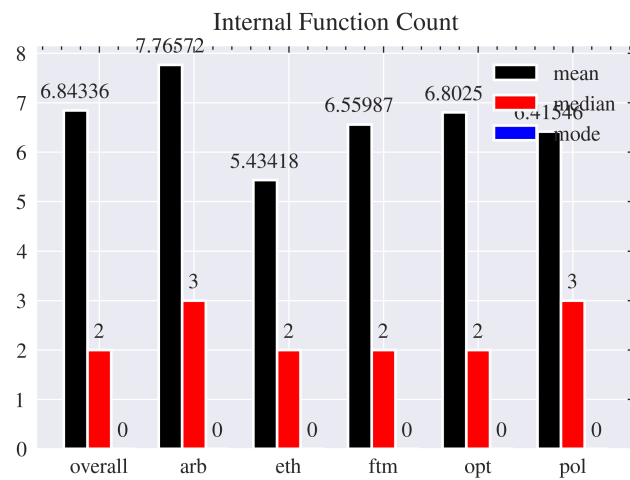
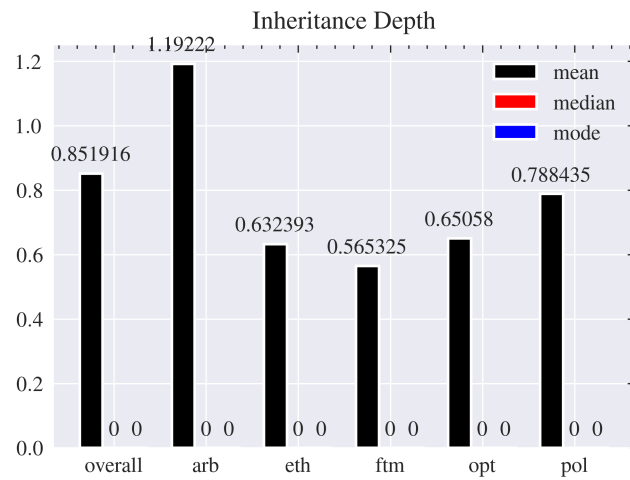
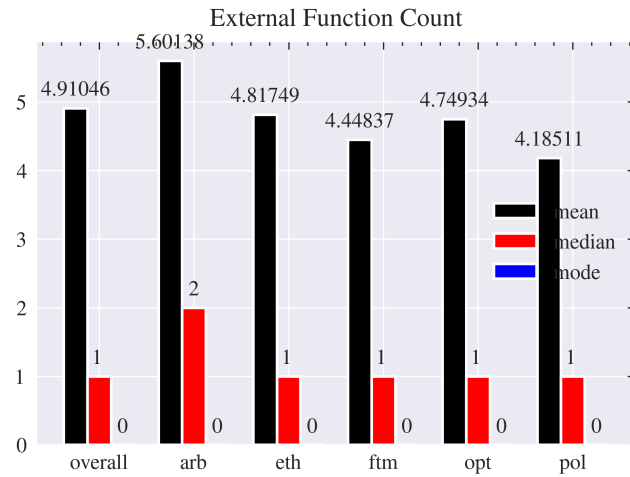
    ax.set_xticks(label_loc + width, labels)

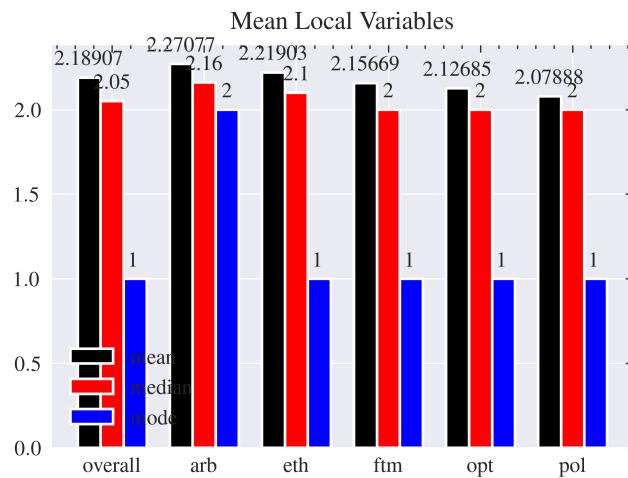
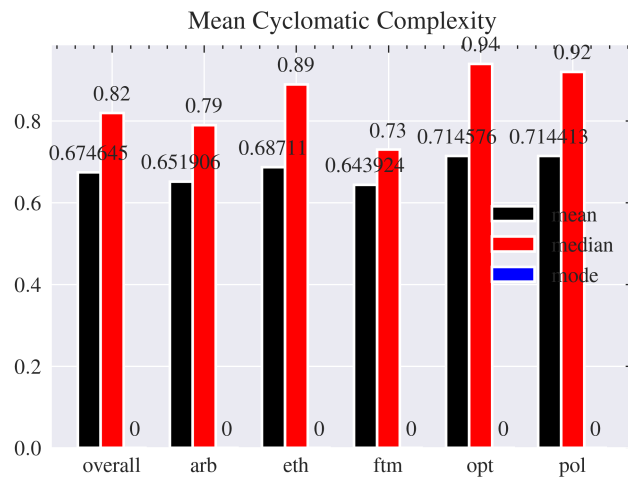
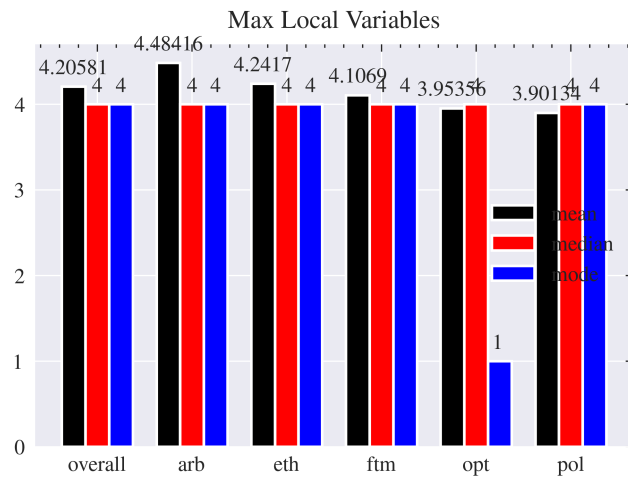
    plt.title(title)
    plt.legend()
    plt.show()

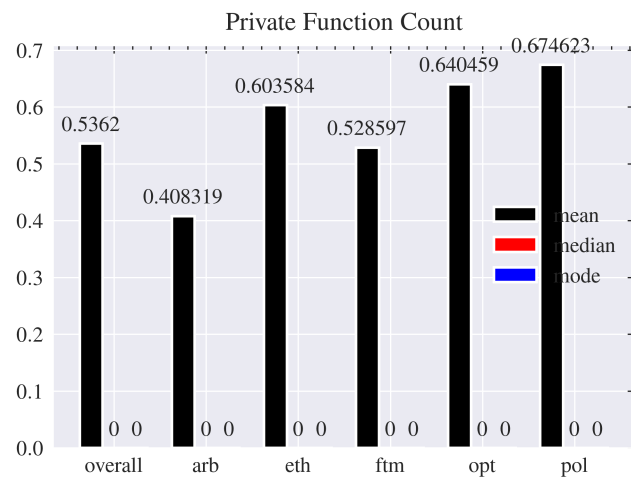
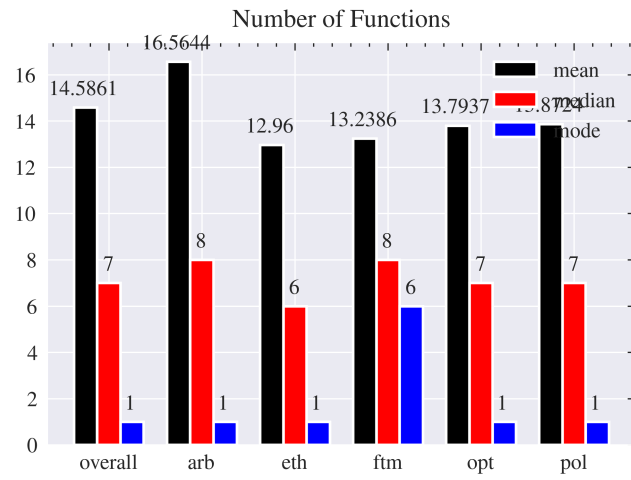
for metric in combined_stats.columns:
    plot_averages(combined_metrics_df, metric, metric)

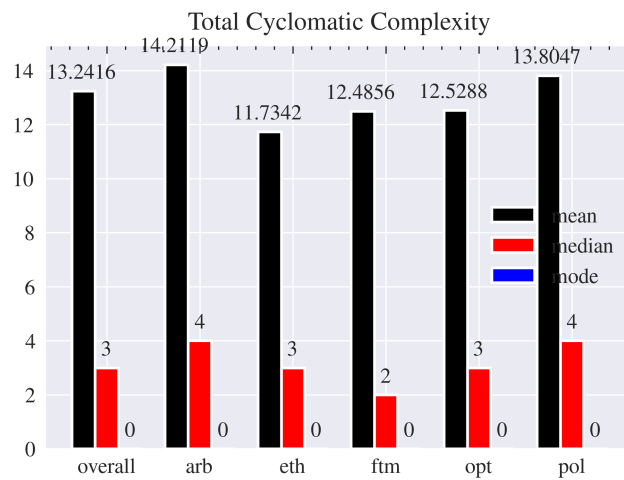
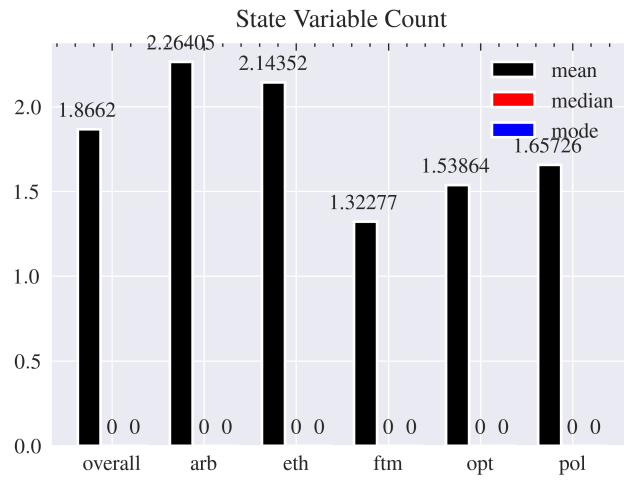
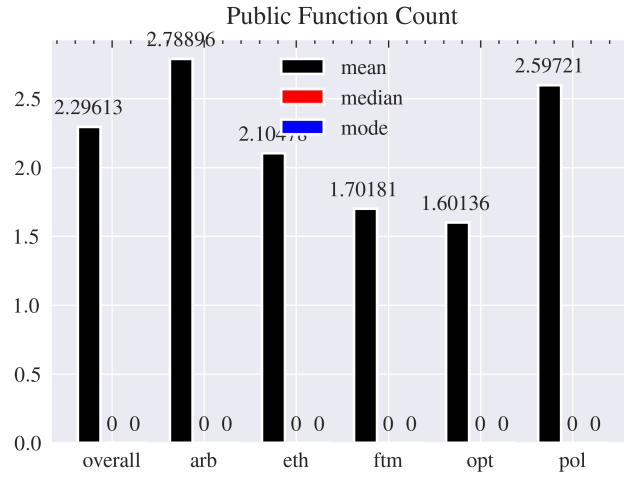
```











## 1.4 Correlation Analysis

### 1.4.1 Setup

```
[33]: metrics = combined_metrics_df.columns.drop('Blockchain')
patterns = combined_patterns_df.columns.drop('Blockchain')
# drop patterns removed from the final analysis
patterns = patterns.drop('Multi Return')
patterns = patterns.drop('State Machine')
patterns = patterns.drop('Permissioned')
patterns = patterns.drop('Permissionless')
```

### 1.4.2 Spearman

```
[34]: def spearman_corr(blockchain=None):
    """
    Computes Spearman correlation between software metrics and micropatterns.
    If a blockchain is specified, it computes the correlation for that
    ↪blockchain only.
    """
    corr_matrix = pd.DataFrame(index=metrics, columns=patterns)

    title = "Spearman Correlation Between Software Metrics and Micro-patterns"

    if blockchain:
        blockchain_metrics_df = ↪
        ↪combined_metrics_df[combined_metrics_df['Blockchain'] == blockchain]
        blockchain_patterns_df = ↪
        ↪combined_patterns_df[combined_patterns_df['Blockchain'] == blockchain]
        title = f"{title} ({blockchain})"
        print(title)
    else:
        blockchain_metrics_df = combined_metrics_df
        blockchain_patterns_df = combined_patterns_df
        title = f"{title} (overall)"
        print(title)

    for pattern in patterns:
        for metric in metrics:
            r, p = stats.spearmanr(blockchain_metrics_df[metric], ↪
            ↪blockchain_patterns_df[pattern])
            corr_matrix.loc[metric, pattern] = r

    corr_matrix = corr_matrix.astype(float)
    print(corr_matrix)
```

```

# Plot correlation matrix
plt.figure(figsize=(10, 6))
sns.heatmap(corr_matrix, annot=True, fmt=".2f", cmap="coolwarm", center=0,
linewidths=0.5)
plt.title(title)
plt.xlabel("Micro-patterns")
plt.ylabel("Software Metrics")
plt.show()

# Compute Spearman correlation for overall and per blockchain
spearman_corr()
for blockchain in blockchains:
    spearman_corr(blockchain)

```

#### Spearman Correlation Between Software Metrics and Micro-patterns (overall)

|                             | Borrower | Delegator | Emitter   | Implementer \ |
|-----------------------------|----------|-----------|-----------|---------------|
| Contract Lines of Code      | 0.278062 | 0.364430  | -0.063355 | 0.057212      |
| External Function Count     | 0.115751 | 0.065180  | -0.151784 | 0.601377      |
| Inheritance Depth           | 0.463816 | 0.412185  | -0.107958 | 0.813823      |
| Internal Function Count     | 0.567611 | 0.610849  | -0.136961 | 0.234617      |
| Max Local Variables         | 0.330217 | 0.465963  | -0.220550 | 0.280125      |
| Mean Cyclomatic Complexity  | 0.433625 | 0.522360  | -0.118990 | -0.022557     |
| Mean Local Variables        | 0.046123 | 0.199657  | -0.211973 | 0.016799      |
| Number of Functions         | 0.463603 | 0.503004  | -0.228209 | 0.520812      |
| Private Function Count      | 0.366864 | 0.470194  | -0.067362 | 0.180001      |
| Public Function Count       | 0.486667 | 0.495961  | -0.087125 | 0.521586      |
| State Variable Count        | 0.572554 | 0.514398  | -0.088055 | 0.364319      |
| Total Cyclomatic Complexity | 0.592579 | 0.658316  | -0.137163 | 0.286640      |

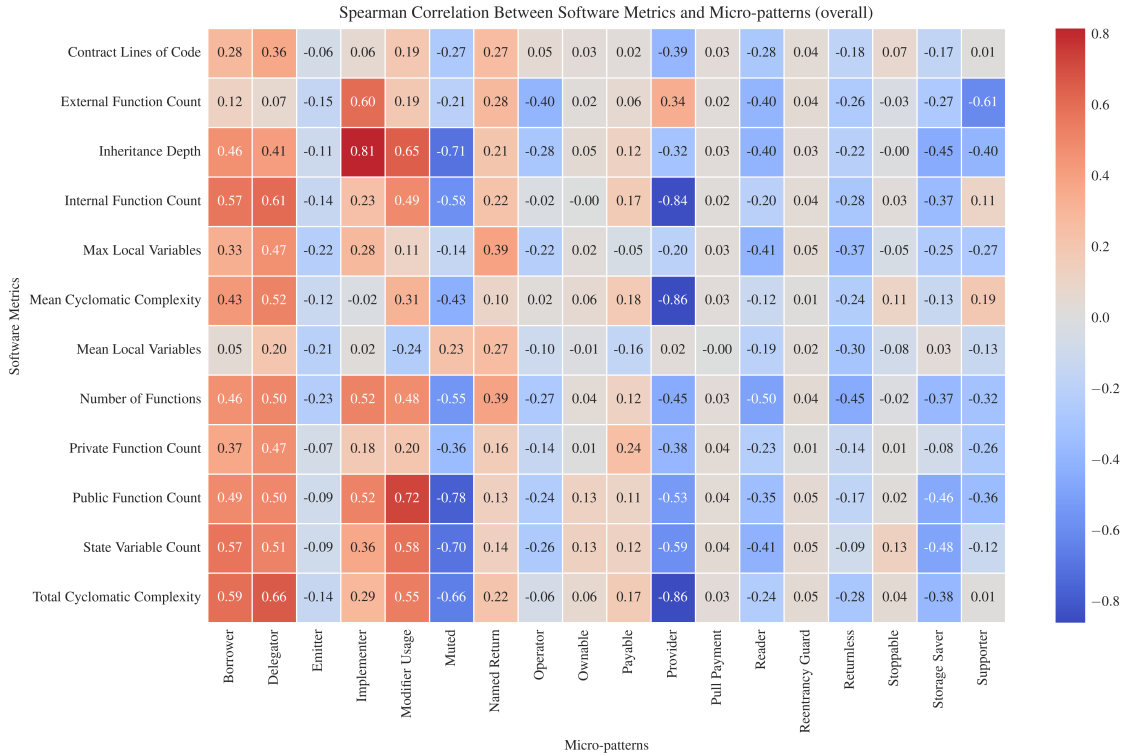
|                             | Modifier Usage | Muted     | Named Return | Operator \ |
|-----------------------------|----------------|-----------|--------------|------------|
| Contract Lines of Code      | 0.194401       | -0.265618 | 0.267472     | 0.048486   |
| External Function Count     | 0.191885       | -0.205593 | 0.283005     | -0.399802  |
| Inheritance Depth           | 0.647639       | -0.707397 | 0.207288     | -0.283269  |
| Internal Function Count     | 0.490079       | -0.584123 | 0.223016     | -0.019487  |
| Max Local Variables         | 0.110940       | -0.144065 | 0.390301     | -0.222376  |
| Mean Cyclomatic Complexity  | 0.311788       | -0.434874 | 0.095797     | 0.017321   |
| Mean Local Variables        | -0.239199      | 0.230695  | 0.268976     | -0.098162  |
| Number of Functions         | 0.478253       | -0.552230 | 0.387671     | -0.269367  |
| Private Function Count      | 0.199748       | -0.362742 | 0.163521     | -0.139527  |
| Public Function Count       | 0.723545       | -0.779081 | 0.133268     | -0.243291  |
| State Variable Count        | 0.578633       | -0.698716 | 0.139349     | -0.262533  |
| Total Cyclomatic Complexity | 0.545472       | -0.658150 | 0.217518     | -0.062332  |

|                         | Ownable  | Payable  | Provider  | Pull Payment \ |
|-------------------------|----------|----------|-----------|----------------|
| Contract Lines of Code  | 0.029302 | 0.024328 | -0.385155 | 0.028807       |
| External Function Count | 0.020523 | 0.055152 | 0.340134  | 0.024736       |
| Inheritance Depth       | 0.047103 | 0.120293 | -0.318219 | 0.027613       |

|                             |           |           |           |           |
|-----------------------------|-----------|-----------|-----------|-----------|
| Internal Function Count     | -0.003005 | 0.171870  | -0.844661 | 0.019879  |
| Max Local Variables         | 0.024819  | -0.045681 | -0.197501 | 0.025909  |
| Mean Cyclomatic Complexity  | 0.055526  | 0.176625  | -0.861412 | 0.025700  |
| Mean Local Variables        | -0.013265 | -0.159958 | 0.024142  | -0.000409 |
| Number of Functions         | 0.040039  | 0.124097  | -0.449968 | 0.030369  |
| Private Function Count      | 0.007905  | 0.243873  | -0.384315 | 0.035534  |
| Public Function Count       | 0.125431  | 0.105244  | -0.529477 | 0.041588  |
| State Variable Count        | 0.130471  | 0.121736  | -0.587446 | 0.042751  |
| Total Cyclomatic Complexity | 0.061693  | 0.168999  | -0.861308 | 0.033160  |

|                             | Reader    | Reentrancy Guard | Returnless \ |
|-----------------------------|-----------|------------------|--------------|
| Contract Lines of Code      | -0.278727 | 0.040117         | -0.178113    |
| External Function Count     | -0.400077 | 0.044008         | -0.256321    |
| Inheritance Depth           | -0.403823 | 0.033532         | -0.218381    |
| Internal Function Count     | -0.203752 | 0.038251         | -0.281239    |
| Max Local Variables         | -0.405614 | 0.046693         | -0.367770    |
| Mean Cyclomatic Complexity  | -0.120495 | 0.013712         | -0.236762    |
| Mean Local Variables        | -0.194493 | 0.024212         | -0.302739    |
| Number of Functions         | -0.498505 | 0.042979         | -0.454338    |
| Private Function Count      | -0.226356 | 0.012846         | -0.139480    |
| Public Function Count       | -0.346307 | 0.048410         | -0.168746    |
| State Variable Count        | -0.413504 | 0.054530         | -0.087317    |
| Total Cyclomatic Complexity | -0.242662 | 0.045686         | -0.279737    |

|                             | Stoppable | Storage Saver | Supporter |
|-----------------------------|-----------|---------------|-----------|
| Contract Lines of Code      | 0.071852  | -0.167575     | 0.009376  |
| External Function Count     | -0.033110 | -0.271887     | -0.608181 |
| Inheritance Depth           | -0.004652 | -0.450320     | -0.399345 |
| Internal Function Count     | 0.025927  | -0.367546     | 0.106780  |
| Max Local Variables         | -0.048621 | -0.247405     | -0.269673 |
| Mean Cyclomatic Complexity  | 0.105207  | -0.130529     | 0.190868  |
| Mean Local Variables        | -0.080242 | 0.031642      | -0.131287 |
| Number of Functions         | -0.016941 | -0.373429     | -0.319268 |
| Private Function Count      | 0.011998  | -0.080452     | -0.261681 |
| Public Function Count       | 0.019005  | -0.459734     | -0.360521 |
| State Variable Count        | 0.126886  | -0.484609     | -0.116699 |
| Total Cyclomatic Complexity | 0.043997  | -0.382603     | 0.009888  |



Spearman Correlation Between Software Metrics and Micro-patterns (arb)

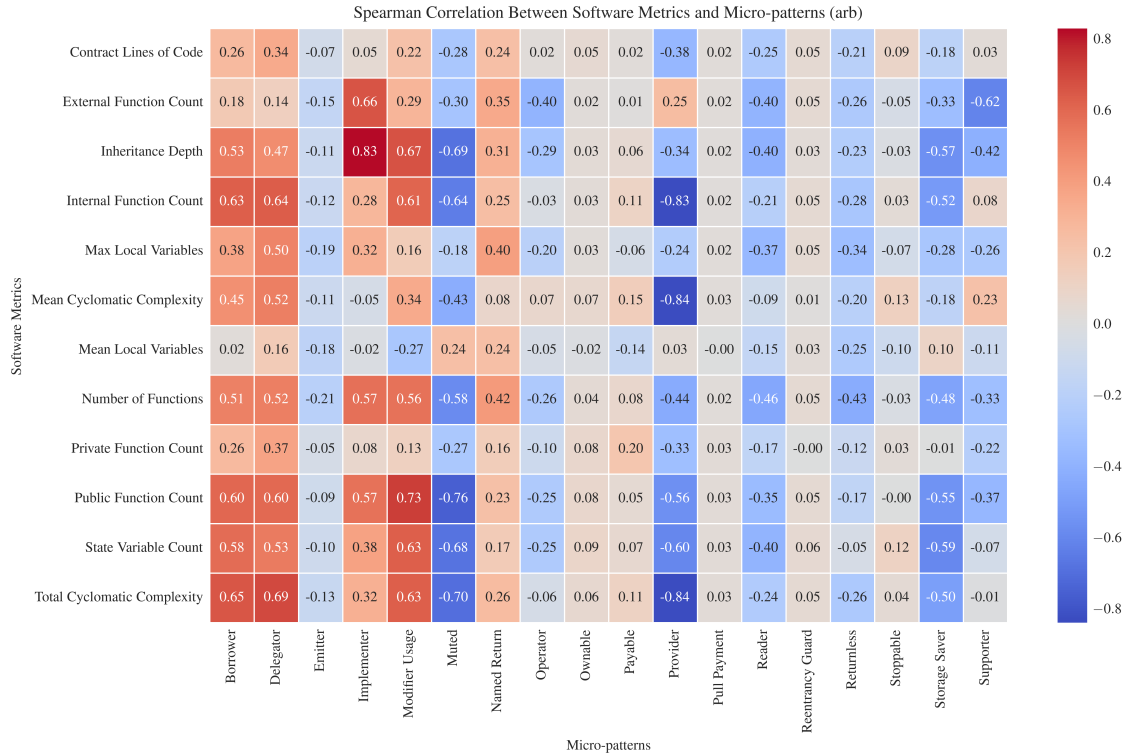
|                             | Borrower       | Delegator | Emitter      | Implementer | \ |
|-----------------------------|----------------|-----------|--------------|-------------|---|
| Contract Lines of Code      | 0.264009       | 0.340566  | -0.071904    | 0.048716    |   |
| External Function Count     | 0.176597       | 0.136664  | -0.149969    | 0.662313    |   |
| Inheritance Depth           | 0.526011       | 0.465810  | -0.111130    | 0.827811    |   |
| Internal Function Count     | 0.628307       | 0.635620  | -0.120997    | 0.281960    |   |
| Max Local Variables         | 0.378088       | 0.495592  | -0.193720    | 0.317400    |   |
| Mean Cyclomatic Complexity  | 0.451957       | 0.520488  | -0.106235    | -0.054438   |   |
| Mean Local Variables        | 0.020204       | 0.163310  | -0.176489    | -0.020002   |   |
| Number of Functions         | 0.508892       | 0.520393  | -0.206122    | 0.567782    |   |
| Private Function Count      | 0.261329       | 0.372677  | -0.053947    | 0.078329    |   |
| Public Function Count       | 0.603745       | 0.597783  | -0.090802    | 0.565062    |   |
| State Variable Count        | 0.583362       | 0.532784  | -0.097185    | 0.383581    |   |
| Total Cyclomatic Complexity | 0.654563       | 0.691231  | -0.125643    | 0.318442    |   |
|                             | Modifier Usage | Muted     | Named Return | Operator    | \ |
| Contract Lines of Code      | 0.224053       | -0.277294 | 0.238401     | 0.022157    |   |
| External Function Count     | 0.292391       | -0.297943 | 0.348103     | -0.396682   |   |
| Inheritance Depth           | 0.671442       | -0.687297 | 0.307950     | -0.290134   |   |
| Internal Function Count     | 0.608797       | -0.640006 | 0.246419     | -0.028208   |   |
| Max Local Variables         | 0.163034       | -0.182232 | 0.396422     | -0.195221   |   |
| Mean Cyclomatic Complexity  | 0.343911       | -0.427737 | 0.079213     | 0.072785    |   |
| Mean Local Variables        | -0.274572      | 0.241211  | 0.239074     | -0.049638   |   |

|                             |          |           |          |           |
|-----------------------------|----------|-----------|----------|-----------|
| Number of Functions         | 0.564519 | -0.584424 | 0.415265 | -0.256595 |
| Private Function Count      | 0.130587 | -0.270614 | 0.156158 | -0.099496 |
| Public Function Count       | 0.727853 | -0.758035 | 0.233089 | -0.251651 |
| State Variable Count        | 0.627762 | -0.677827 | 0.168973 | -0.252503 |
| Total Cyclomatic Complexity | 0.631141 | -0.698782 | 0.256075 | -0.056391 |

|                             | Ownable   | Payable   | Provider  | Pull Payment \ |
|-----------------------------|-----------|-----------|-----------|----------------|
| Contract Lines of Code      | 0.053619  | 0.024771  | -0.377406 | 0.024977       |
| External Function Count     | 0.020111  | 0.009172  | 0.254159  | 0.020674       |
| Inheritance Depth           | 0.033279  | 0.058567  | -0.338428 | 0.019403       |
| Internal Function Count     | 0.026843  | 0.110540  | -0.829321 | 0.018502       |
| Max Local Variables         | 0.026248  | -0.057651 | -0.238063 | 0.023471       |
| Mean Cyclomatic Complexity  | 0.069309  | 0.148424  | -0.839020 | 0.026250       |
| Mean Local Variables        | -0.015790 | -0.137885 | 0.034629  | -0.002829      |
| Number of Functions         | 0.044603  | 0.075347  | -0.438550 | 0.024666       |
| Private Function Count      | 0.078019  | 0.198667  | -0.328127 | 0.027317       |
| Public Function Count       | 0.075345  | 0.048634  | -0.561063 | 0.030850       |
| State Variable Count        | 0.086098  | 0.065426  | -0.604895 | 0.032740       |
| Total Cyclomatic Complexity | 0.059956  | 0.112692  | -0.840106 | 0.028256       |

|                             | Reader    | Reentrancy Guard | Returnless \ |
|-----------------------------|-----------|------------------|--------------|
| Contract Lines of Code      | -0.249503 | 0.050675         | -0.207068    |
| External Function Count     | -0.395331 | 0.047809         | -0.258337    |
| Inheritance Depth           | -0.403650 | 0.027832         | -0.228329    |
| Internal Function Count     | -0.214104 | 0.048250         | -0.280140    |
| Max Local Variables         | -0.370821 | 0.051225         | -0.336796    |
| Mean Cyclomatic Complexity  | -0.086948 | 0.007889         | -0.202659    |
| Mean Local Variables        | -0.153867 | 0.031404         | -0.252471    |
| Number of Functions         | -0.456058 | 0.049663         | -0.431075    |
| Private Function Count      | -0.166641 | -0.003977        | -0.117323    |
| Public Function Count       | -0.349274 | 0.050740         | -0.166161    |
| State Variable Count        | -0.396633 | 0.057035         | -0.051222    |
| Total Cyclomatic Complexity | -0.241832 | 0.050527         | -0.263779    |

|                             | Stoppable | Storage Saver | Supporter |
|-----------------------------|-----------|---------------|-----------|
| Contract Lines of Code      | 0.094812  | -0.184293     | 0.029333  |
| External Function Count     | -0.049021 | -0.334904     | -0.615418 |
| Inheritance Depth           | -0.025023 | -0.565224     | -0.415541 |
| Internal Function Count     | 0.034083  | -0.515774     | 0.081107  |
| Max Local Variables         | -0.066506 | -0.284771     | -0.261956 |
| Mean Cyclomatic Complexity  | 0.126916  | -0.177756     | 0.226804  |
| Mean Local Variables        | -0.099090 | 0.101185      | -0.105438 |
| Number of Functions         | -0.026459 | -0.478432     | -0.328046 |
| Private Function Count      | 0.027651  | -0.005438     | -0.218555 |
| Public Function Count       | -0.003498 | -0.552763     | -0.373707 |
| State Variable Count        | 0.116926  | -0.594609     | -0.071209 |
| Total Cyclomatic Complexity | 0.042660  | -0.496117     | -0.011611 |



Spearman Correlation Between Software Metrics and Micro-patterns (eth)

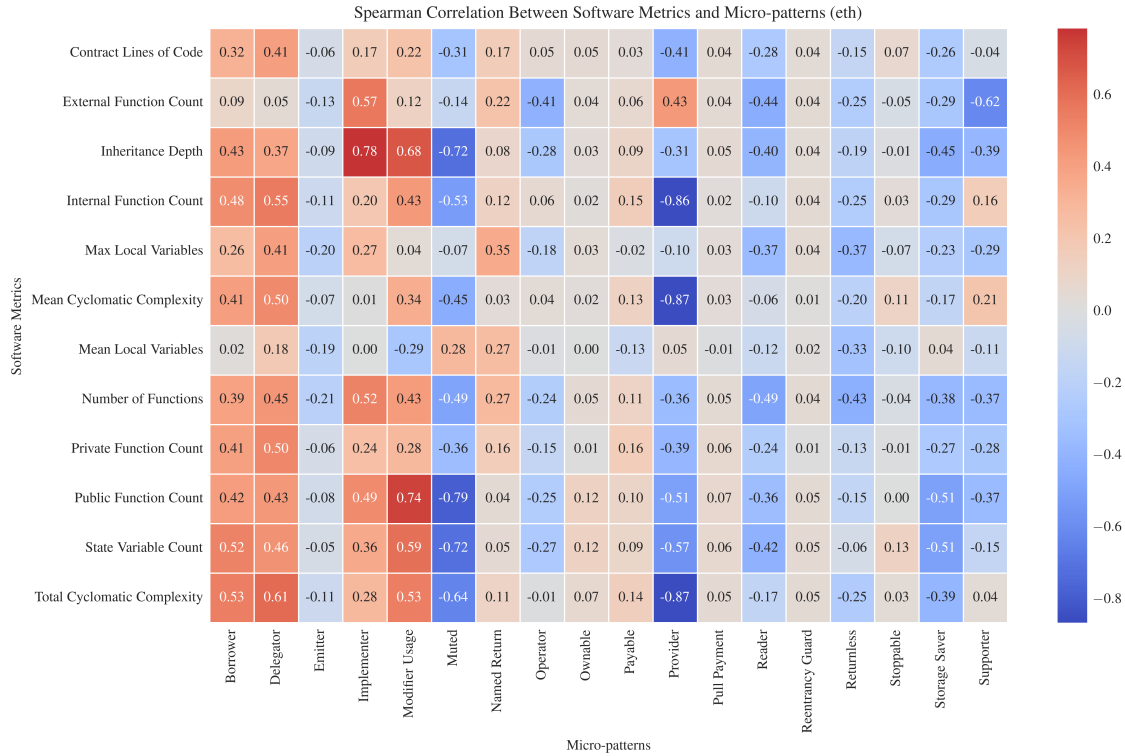
|                             | Borrower       | Delegator | Emitter      | Implementer | \ |
|-----------------------------|----------------|-----------|--------------|-------------|---|
| Contract Lines of Code      | 0.323886       | 0.411885  | -0.059528    | 0.168824    |   |
| External Function Count     | 0.089831       | 0.045175  | -0.132611    | 0.569443    |   |
| Inheritance Depth           | 0.426110       | 0.371675  | -0.091304    | 0.782538    |   |
| Internal Function Count     | 0.483705       | 0.550450  | -0.110457    | 0.201115    |   |
| Max Local Variables         | 0.264832       | 0.414255  | -0.200820    | 0.267417    |   |
| Mean Cyclomatic Complexity  | 0.411912       | 0.497905  | -0.070962    | 0.006631    |   |
| Mean Local Variables        | 0.017863       | 0.183447  | -0.191965    | 0.000251    |   |
| Number of Functions         | 0.393277       | 0.448222  | -0.205374    | 0.523032    |   |
| Private Function Count      | 0.405574       | 0.501901  | -0.061484    | 0.241653    |   |
| Public Function Count       | 0.418721       | 0.434284  | -0.080025    | 0.488535    |   |
| State Variable Count        | 0.522828       | 0.463141  | -0.051146    | 0.357703    |   |
| Total Cyclomatic Complexity | 0.533245       | 0.611661  | -0.109948    | 0.284798    |   |
|                             | Modifier Usage | Muted     | Named Return | Operator    | \ |
| Contract Lines of Code      | 0.222418       | -0.307981 | 0.165144     | 0.050771    |   |
| External Function Count     | 0.118361       | -0.135062 | 0.224498     | -0.413487   |   |
| Inheritance Depth           | 0.678479       | -0.721174 | 0.081432     | -0.278306   |   |
| Internal Function Count     | 0.432264       | -0.530321 | 0.120060     | 0.059778    |   |
| Max Local Variables         | 0.039750       | -0.070123 | 0.350716     | -0.179952   |   |
| Mean Cyclomatic Complexity  | 0.340297       | -0.445921 | 0.030785     | 0.038120    |   |
| Mean Local Variables        | -0.293320      | 0.280144  | 0.265053     | -0.011642   |   |

|                             |          |           |          |           |
|-----------------------------|----------|-----------|----------|-----------|
| Number of Functions         | 0.425756 | -0.493705 | 0.274273 | -0.240606 |
| Private Function Count      | 0.284903 | -0.355419 | 0.158242 | -0.150447 |
| Public Function Count       | 0.741713 | -0.794393 | 0.044061 | -0.245935 |
| State Variable Count        | 0.593293 | -0.715543 | 0.050420 | -0.273278 |
| Total Cyclomatic Complexity | 0.532344 | -0.637121 | 0.108526 | -0.010531 |

|                             | Ownable  | Payable   | Provider  | Pull Payment \ |
|-----------------------------|----------|-----------|-----------|----------------|
| Contract Lines of Code      | 0.047688 | 0.033240  | -0.414822 | 0.044652       |
| External Function Count     | 0.036515 | 0.060785  | 0.430334  | 0.041893       |
| Inheritance Depth           | 0.031309 | 0.092612  | -0.310994 | 0.049299       |
| Internal Function Count     | 0.017390 | 0.152128  | -0.859458 | 0.021601       |
| Max Local Variables         | 0.034505 | -0.024622 | -0.098319 | 0.033066       |
| Mean Cyclomatic Complexity  | 0.022445 | 0.128764  | -0.868614 | 0.033127       |
| Mean Local Variables        | 0.000948 | -0.126608 | 0.050381  | -0.008264      |
| Number of Functions         | 0.050912 | 0.105362  | -0.359280 | 0.047046       |
| Private Function Count      | 0.007628 | 0.158190  | -0.388274 | 0.060314       |
| Public Function Count       | 0.119195 | 0.095291  | -0.510315 | 0.067130       |
| State Variable Count        | 0.116671 | 0.088513  | -0.570206 | 0.064897       |
| Total Cyclomatic Complexity | 0.065535 | 0.144168  | -0.866210 | 0.050217       |

|                             | Reader    | Reentrancy Guard | Returnless \ |
|-----------------------------|-----------|------------------|--------------|
| Contract Lines of Code      | -0.275206 | 0.035799         | -0.150307    |
| External Function Count     | -0.438681 | 0.041577         | -0.245083    |
| Inheritance Depth           | -0.402798 | 0.039728         | -0.186895    |
| Internal Function Count     | -0.101511 | 0.041051         | -0.245320    |
| Max Local Variables         | -0.366782 | 0.044233         | -0.372989    |
| Mean Cyclomatic Complexity  | -0.056378 | 0.010566         | -0.203186    |
| Mean Local Variables        | -0.118288 | 0.020214         | -0.325374    |
| Number of Functions         | -0.489668 | 0.044051         | -0.431527    |
| Private Function Count      | -0.239089 | 0.005210         | -0.131672    |
| Public Function Count       | -0.357396 | 0.046950         | -0.154456    |
| State Variable Count        | -0.422338 | 0.052965         | -0.064929    |
| Total Cyclomatic Complexity | -0.168676 | 0.045619         | -0.253719    |

|                             | Stoppable | Storage Saver | Supporter |
|-----------------------------|-----------|---------------|-----------|
| Contract Lines of Code      | 0.072790  | -0.256746     | -0.044105 |
| External Function Count     | -0.045406 | -0.286082     | -0.623429 |
| Inheritance Depth           | -0.012092 | -0.454725     | -0.389600 |
| Internal Function Count     | 0.031674  | -0.291226     | 0.159814  |
| Max Local Variables         | -0.072844 | -0.234952     | -0.294458 |
| Mean Cyclomatic Complexity  | 0.106981  | -0.168626     | 0.213453  |
| Mean Local Variables        | -0.100975 | 0.043786      | -0.114726 |
| Number of Functions         | -0.041943 | -0.382681     | -0.367628 |
| Private Function Count      | -0.009944 | -0.272026     | -0.280972 |
| Public Function Count       | 0.000009  | -0.509147     | -0.369286 |
| State Variable Count        | 0.126627  | -0.511864     | -0.150940 |
| Total Cyclomatic Complexity | 0.034682  | -0.393210     | 0.037897  |



Spearman Correlation Between Software Metrics and Micro-patterns (ftm)

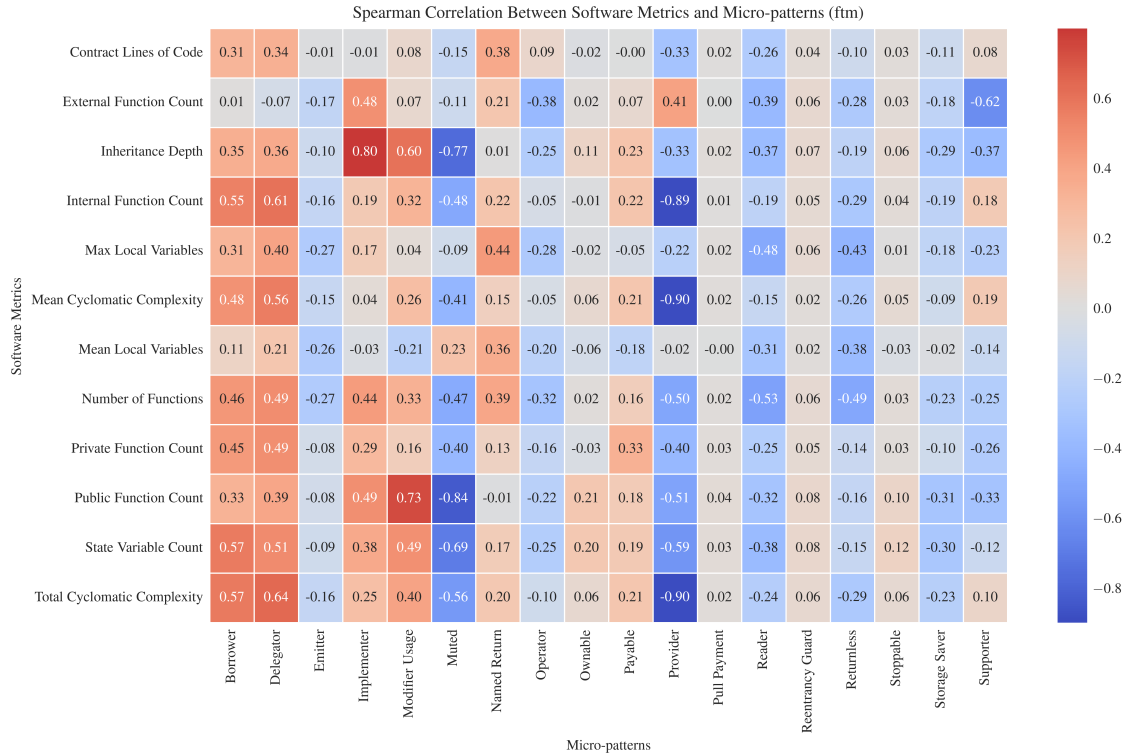
|                             | Borrower       | Delegator | Emitter      | Implementer | \ |
|-----------------------------|----------------|-----------|--------------|-------------|---|
| Contract Lines of Code      | 0.308473       | 0.340585  | -0.011086    | -0.010660   |   |
| External Function Count     | 0.014811       | -0.069115 | -0.171966    | 0.481185    |   |
| Inheritance Depth           | 0.345296       | 0.361653  | -0.104215    | 0.798973    |   |
| Internal Function Count     | 0.550291       | 0.606560  | -0.161512    | 0.188876    |   |
| Max Local Variables         | 0.311658       | 0.403119  | -0.267936    | 0.172845    |   |
| Mean Cyclomatic Complexity  | 0.478799       | 0.561402  | -0.147719    | 0.042501    |   |
| Mean Local Variables        | 0.111708       | 0.209359  | -0.263690    | -0.026992   |   |
| Number of Functions         | 0.460109       | 0.494422  | -0.268186    | 0.439123    |   |
| Private Function Count      | 0.447048       | 0.493210  | -0.075551    | 0.290704    |   |
| Public Function Count       | 0.326709       | 0.391109  | -0.081039    | 0.489479    |   |
| State Variable Count        | 0.568377       | 0.512559  | -0.091480    | 0.377238    |   |
| Total Cyclomatic Complexity | 0.573885       | 0.643008  | -0.156830    | 0.254988    |   |
|                             | Modifier Usage | Muted     | Named Return | Operator    | \ |
| Contract Lines of Code      | 0.080617       | -0.150948 | 0.379957     | 0.086269    |   |
| External Function Count     | 0.071941       | -0.105219 | 0.214481     | -0.383040   |   |
| Inheritance Depth           | 0.603802       | -0.767102 | 0.007473     | -0.251230   |   |
| Internal Function Count     | 0.318651       | -0.481495 | 0.221045     | -0.052900   |   |
| Max Local Variables         | 0.038476       | -0.086258 | 0.441275     | -0.276191   |   |
| Mean Cyclomatic Complexity  | 0.262090       | -0.411348 | 0.147233     | -0.047618   |   |
| Mean Local Variables        | -0.213451      | 0.229759  | 0.356590     | -0.203212   |   |

|                             |          |           |           |           |
|-----------------------------|----------|-----------|-----------|-----------|
| Number of Functions         | 0.333946 | -0.474269 | 0.391376  | -0.317752 |
| Private Function Count      | 0.158762 | -0.401149 | 0.131057  | -0.159700 |
| Public Function Count       | 0.734478 | -0.837480 | -0.011486 | -0.216707 |
| State Variable Count        | 0.485516 | -0.693809 | 0.174509  | -0.247453 |
| Total Cyclomatic Complexity | 0.403369 | -0.563499 | 0.199030  | -0.095709 |

|                             | Ownable   | Payable   | Provider  | Pull Payment \ |
|-----------------------------|-----------|-----------|-----------|----------------|
| Contract Lines of Code      | -0.021209 | -0.003956 | -0.327769 | 0.015428       |
| External Function Count     | 0.016868  | 0.065782  | 0.412340  | 0.000788       |
| Inheritance Depth           | 0.108816  | 0.228375  | -0.333469 | 0.023354       |
| Internal Function Count     | -0.013101 | 0.216871  | -0.886638 | 0.012034       |
| Max Local Variables         | -0.019442 | -0.051037 | -0.217404 | 0.017199       |
| Mean Cyclomatic Complexity  | 0.063687  | 0.211888  | -0.899357 | 0.019129       |
| Mean Local Variables        | -0.064620 | -0.184972 | -0.016177 | -0.003659      |
| Number of Functions         | 0.021003  | 0.156585  | -0.495362 | 0.022554       |
| Private Function Count      | -0.033082 | 0.328679  | -0.399145 | 0.025259       |
| Public Function Count       | 0.205290  | 0.179212  | -0.509483 | 0.037476       |
| State Variable Count        | 0.201889  | 0.193520  | -0.586161 | 0.034979       |
| Total Cyclomatic Complexity | 0.063555  | 0.213381  | -0.898631 | 0.023747       |

|                             | Reader    | Reentrancy Guard | Returnless \ |
|-----------------------------|-----------|------------------|--------------|
| Contract Lines of Code      | -0.260723 | 0.040985         | -0.099657    |
| External Function Count     | -0.391113 | 0.064529         | -0.279342    |
| Inheritance Depth           | -0.371794 | 0.067665         | -0.194622    |
| Internal Function Count     | -0.193631 | 0.053668         | -0.291752    |
| Max Local Variables         | -0.475068 | 0.061226         | -0.425349    |
| Mean Cyclomatic Complexity  | -0.146708 | 0.023286         | -0.257134    |
| Mean Local Variables        | -0.314270 | 0.021599         | -0.380743    |
| Number of Functions         | -0.527452 | 0.060950         | -0.490783    |
| Private Function Count      | -0.247214 | 0.053202         | -0.136362    |
| Public Function Count       | -0.316344 | 0.075977         | -0.159427    |
| State Variable Count        | -0.379549 | 0.080273         | -0.147315    |
| Total Cyclomatic Complexity | -0.236494 | 0.062797         | -0.287844    |

|                             | Stoppable | Storage Saver | Supporter |
|-----------------------------|-----------|---------------|-----------|
| Contract Lines of Code      | 0.029135  | -0.107975     | 0.075069  |
| External Function Count     | 0.030049  | -0.183622     | -0.617210 |
| Inheritance Depth           | 0.061834  | -0.286133     | -0.371308 |
| Internal Function Count     | 0.035794  | -0.193861     | 0.183184  |
| Max Local Variables         | 0.007456  | -0.177248     | -0.230293 |
| Mean Cyclomatic Complexity  | 0.054406  | -0.091474     | 0.191221  |
| Mean Local Variables        | -0.032252 | -0.018945     | -0.140171 |
| Number of Functions         | 0.033228  | -0.225789     | -0.246815 |
| Private Function Count      | 0.025607  | -0.095947     | -0.257059 |
| Public Function Count       | 0.096126  | -0.310066     | -0.328119 |
| State Variable Count        | 0.118691  | -0.302051     | -0.122118 |
| Total Cyclomatic Complexity | 0.062269  | -0.226092     | 0.097899  |



Spearman Correlation Between Software Metrics and Micro-patterns (opt)

|                             | Borrower | Delegator | Emitter   | Implementer | \ |
|-----------------------------|----------|-----------|-----------|-------------|---|
| Contract Lines of Code      | 0.319000 | 0.442117  | -0.132470 | 0.081976    |   |
| External Function Count     | 0.026380 | -0.030393 | -0.158310 | 0.548591    |   |
| Inheritance Depth           | 0.450769 | 0.362508  | -0.111674 | 0.826009    |   |
| Internal Function Count     | 0.558191 | 0.628211  | -0.156457 | 0.220028    |   |
| Max Local Variables         | 0.313214 | 0.494323  | -0.259335 | 0.251924    |   |
| Mean Cyclomatic Complexity  | 0.489693 | 0.584595  | -0.145677 | 0.019381    |   |
| Mean Local Variables        | 0.045010 | 0.260974  | -0.254405 | 0.014425    |   |
| Number of Functions         | 0.435263 | 0.501525  | -0.263765 | 0.503130    |   |
| Private Function Count      | 0.465029 | 0.594318  | -0.087557 | 0.288874    |   |
| Public Function Count       | 0.407564 | 0.388644  | -0.089078 | 0.454322    |   |
| State Variable Count        | 0.594883 | 0.493512  | -0.095299 | 0.349329    |   |
| Total Cyclomatic Complexity | 0.580402 | 0.669211  | -0.158556 | 0.256563    |   |

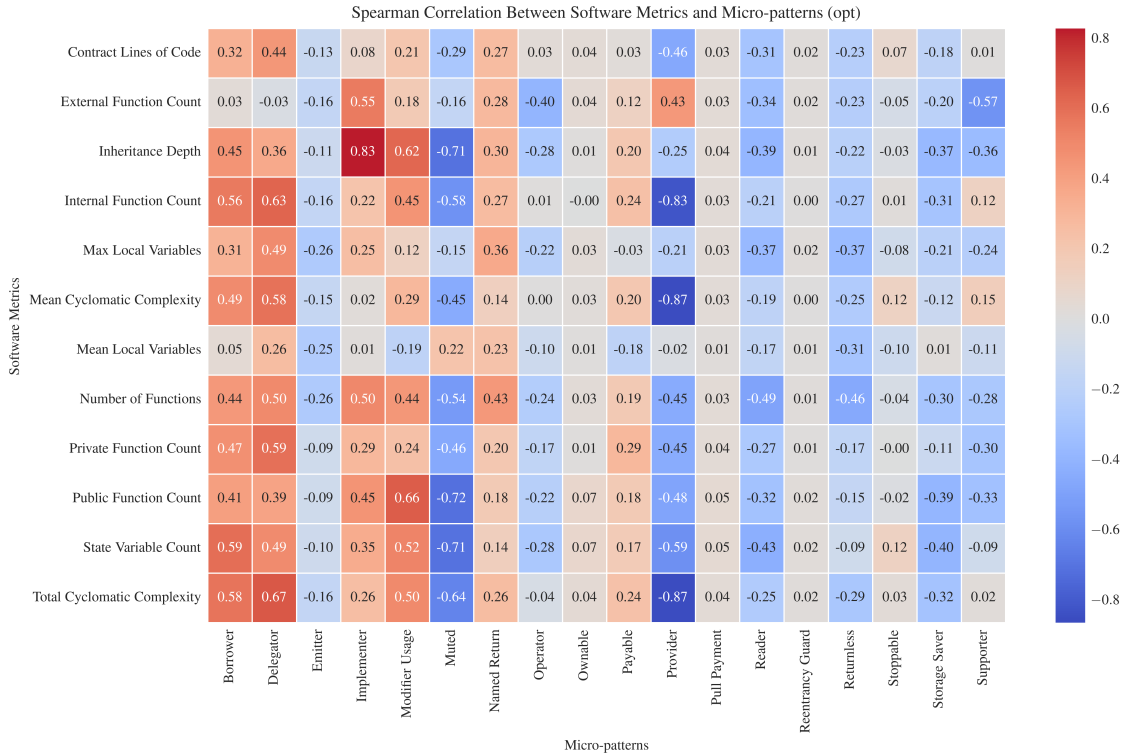
|                            | Modifier Usage | Muted     | Named Return | Operator  | \ |
|----------------------------|----------------|-----------|--------------|-----------|---|
| Contract Lines of Code     | 0.210550       | -0.285078 | 0.265665     | 0.030418  |   |
| External Function Count    | 0.181542       | -0.157607 | 0.279191     | -0.401234 |   |
| Inheritance Depth          | 0.624863       | -0.712328 | 0.303444     | -0.276297 |   |
| Internal Function Count    | 0.452066       | -0.578722 | 0.269094     | 0.008697  |   |
| Max Local Variables        | 0.123661       | -0.147960 | 0.361287     | -0.222693 |   |
| Mean Cyclomatic Complexity | 0.288692       | -0.454955 | 0.136336     | 0.004782  |   |
| Mean Local Variables       | -0.187004      | 0.217962  | 0.229351     | -0.102863 |   |

|                             |          |           |          |           |
|-----------------------------|----------|-----------|----------|-----------|
| Number of Functions         | 0.442083 | -0.537963 | 0.433687 | -0.243764 |
| Private Function Count      | 0.242438 | -0.459700 | 0.201682 | -0.171080 |
| Public Function Count       | 0.659002 | -0.717017 | 0.180713 | -0.224883 |
| State Variable Count        | 0.519103 | -0.708080 | 0.144358 | -0.280557 |
| Total Cyclomatic Complexity | 0.497271 | -0.635638 | 0.258312 | -0.040678 |

|                             | Ownable   | Payable   | Provider  | Pull Payment \ |
|-----------------------------|-----------|-----------|-----------|----------------|
| Contract Lines of Code      | 0.035992  | 0.025606  | -0.463938 | 0.030988       |
| External Function Count     | 0.044818  | 0.120332  | 0.430627  | 0.034347       |
| Inheritance Depth           | 0.011832  | 0.202025  | -0.249572 | 0.036556       |
| Internal Function Count     | -0.002099 | 0.241903  | -0.826147 | 0.030304       |
| Max Local Variables         | 0.031347  | -0.029961 | -0.205931 | 0.031357       |
| Mean Cyclomatic Complexity  | 0.025281  | 0.201400  | -0.866612 | 0.028164       |
| Mean Local Variables        | 0.007682  | -0.180362 | -0.020666 | 0.014114       |
| Number of Functions         | 0.032076  | 0.188787  | -0.454711 | 0.034711       |
| Private Function Count      | 0.006885  | 0.287156  | -0.446116 | 0.035182       |
| Public Function Count       | 0.074709  | 0.181213  | -0.483921 | 0.050992       |
| State Variable Count        | 0.073949  | 0.173193  | -0.588720 | 0.048791       |
| Total Cyclomatic Complexity | 0.039847  | 0.236311  | -0.865893 | 0.037870       |

|                             | Reader    | Reentrancy Guard | Returnless \ |
|-----------------------------|-----------|------------------|--------------|
| Contract Lines of Code      | -0.310595 | 0.015885         | -0.226963    |
| External Function Count     | -0.338128 | 0.015759         | -0.232990    |
| Inheritance Depth           | -0.394455 | 0.013054         | -0.218067    |
| Internal Function Count     | -0.209858 | 0.004649         | -0.269397    |
| Max Local Variables         | -0.372387 | 0.016734         | -0.368964    |
| Mean Cyclomatic Complexity  | -0.187967 | 0.004621         | -0.252397    |
| Mean Local Variables        | -0.165372 | 0.006821         | -0.309522    |
| Number of Functions         | -0.493819 | 0.014792         | -0.458933    |
| Private Function Count      | -0.274243 | 0.006375         | -0.166958    |
| Public Function Count       | -0.322772 | 0.023113         | -0.148983    |
| State Variable Count        | -0.430824 | 0.024643         | -0.087024    |
| Total Cyclomatic Complexity | -0.254653 | 0.016512         | -0.285934    |

|                             | Stoppable | Storage Saver | Supporter |
|-----------------------------|-----------|---------------|-----------|
| Contract Lines of Code      | 0.066856  | -0.183479     | 0.008345  |
| External Function Count     | -0.054881 | -0.200999     | -0.574933 |
| Inheritance Depth           | -0.026537 | -0.374159     | -0.359618 |
| Internal Function Count     | 0.012961  | -0.307588     | 0.121603  |
| Max Local Variables         | -0.079874 | -0.207595     | -0.238360 |
| Mean Cyclomatic Complexity  | 0.117908  | -0.121277     | 0.147358  |
| Mean Local Variables        | -0.103435 | 0.013323      | -0.107993 |
| Number of Functions         | -0.041545 | -0.303505     | -0.281118 |
| Private Function Count      | -0.002476 | -0.113406     | -0.302675 |
| Public Function Count       | -0.015743 | -0.393468     | -0.328325 |
| State Variable Count        | 0.123335  | -0.397270     | -0.091033 |
| Total Cyclomatic Complexity | 0.026715  | -0.316093     | 0.022890  |



Spearman Correlation Between Software Metrics and Micro-patterns (pol)

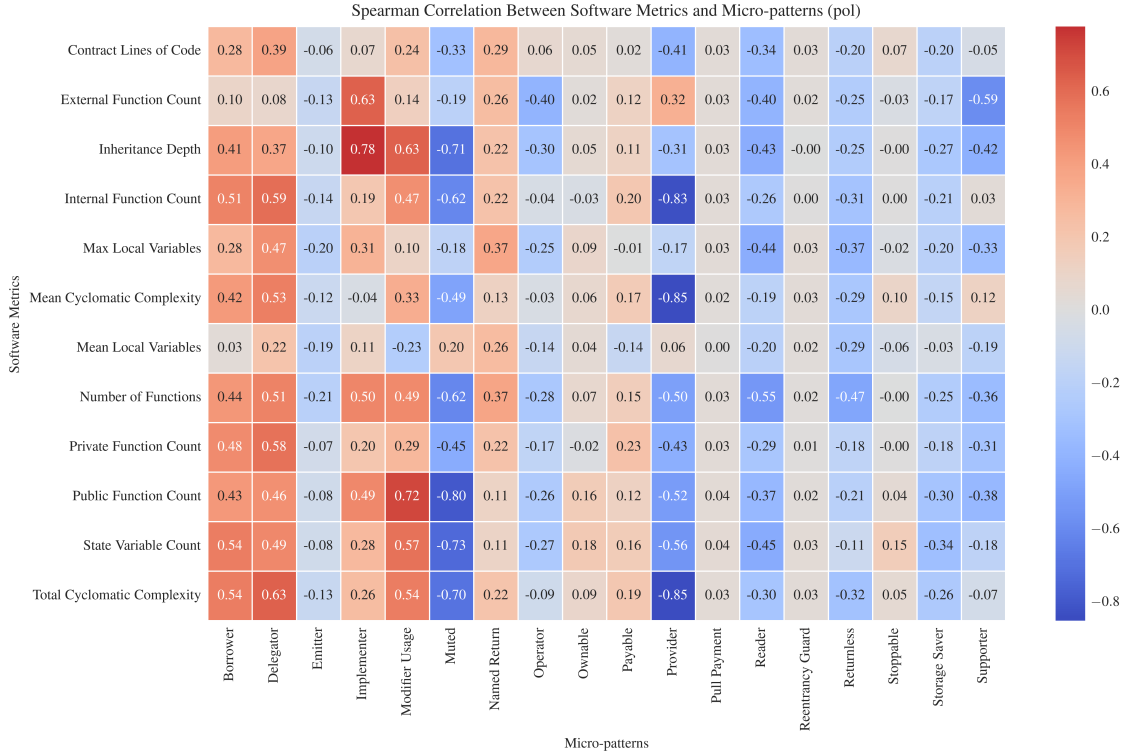
|                             | Borrower       | Delegator | Emitter      | Implementer | \ |
|-----------------------------|----------------|-----------|--------------|-------------|---|
| Contract Lines of Code      | 0.276771       | 0.387246  | -0.064717    | 0.070212    |   |
| External Function Count     | 0.098844       | 0.081021  | -0.134567    | 0.631763    |   |
| Inheritance Depth           | 0.408558       | 0.367254  | -0.104336    | 0.776599    |   |
| Internal Function Count     | 0.513788       | 0.588737  | -0.136014    | 0.191461    |   |
| Max Local Variables         | 0.279047       | 0.470807  | -0.199922    | 0.309141    |   |
| Mean Cyclomatic Complexity  | 0.423562       | 0.530407  | -0.115604    | -0.037165   |   |
| Mean Local Variables        | 0.030496       | 0.215911  | -0.194826    | 0.113602    |   |
| Number of Functions         | 0.436920       | 0.511028  | -0.210413    | 0.501826    |   |
| Private Function Count      | 0.479550       | 0.575350  | -0.073503    | 0.200902    |   |
| Public Function Count       | 0.429045       | 0.459755  | -0.080387    | 0.493699    |   |
| State Variable Count        | 0.536417       | 0.494967  | -0.077305    | 0.275133    |   |
| Total Cyclomatic Complexity | 0.540311       | 0.634788  | -0.133593    | 0.257388    |   |
|                             | Modifier Usage | Muted     | Named Return | Operator    | \ |
| Contract Lines of Code      | 0.235960       | -0.325293 | 0.285943     | 0.062032    |   |
| External Function Count     | 0.135458       | -0.193662 | 0.257613     | -0.403513   |   |
| Inheritance Depth           | 0.634878       | -0.706074 | 0.222859     | -0.302580   |   |
| Internal Function Count     | 0.466959       | -0.616008 | 0.219109     | -0.042366   |   |
| Max Local Variables         | 0.102169       | -0.179816 | 0.367359     | -0.245649   |   |
| Mean Cyclomatic Complexity  | 0.334948       | -0.485124 | 0.125704     | -0.033449   |   |
| Mean Local Variables        | -0.230590      | 0.195585  | 0.262674     | -0.138931   |   |

|                             |          |           |          |           |
|-----------------------------|----------|-----------|----------|-----------|
| Number of Functions         | 0.485351 | -0.616184 | 0.369538 | -0.276915 |
| Private Function Count      | 0.285541 | -0.445690 | 0.220191 | -0.167824 |
| Public Function Count       | 0.719063 | -0.801236 | 0.111998 | -0.258158 |
| State Variable Count        | 0.574833 | -0.734829 | 0.106123 | -0.268243 |
| Total Cyclomatic Complexity | 0.537970 | -0.698574 | 0.215406 | -0.091132 |

|                             | Ownable   | Payable   | Provider  | Pull Payment \ |
|-----------------------------|-----------|-----------|-----------|----------------|
| Contract Lines of Code      | 0.046781  | 0.023849  | -0.408206 | 0.030654       |
| External Function Count     | 0.023504  | 0.118405  | 0.320056  | 0.028009       |
| Inheritance Depth           | 0.049708  | 0.114120  | -0.311612 | 0.025382       |
| Internal Function Count     | -0.034094 | 0.196153  | -0.830286 | 0.025872       |
| Max Local Variables         | 0.085066  | -0.005332 | -0.169612 | 0.028648       |
| Mean Cyclomatic Complexity  | 0.064954  | 0.166744  | -0.853161 | 0.020376       |
| Mean Local Variables        | 0.039700  | -0.137661 | 0.058971  | 0.003591       |
| Number of Functions         | 0.066201  | 0.152124  | -0.501362 | 0.030743       |
| Private Function Count      | -0.018227 | 0.229421  | -0.429789 | 0.031100       |
| Public Function Count       | 0.162533  | 0.122089  | -0.522339 | 0.036290       |
| State Variable Count        | 0.181733  | 0.162677  | -0.556481 | 0.042503       |
| Total Cyclomatic Complexity | 0.086807  | 0.190651  | -0.854162 | 0.032285       |

|                             | Reader    | Reentrancy Guard | Returnless \ |
|-----------------------------|-----------|------------------|--------------|
| Contract Lines of Code      | -0.339339 | 0.026846         | -0.203663    |
| External Function Count     | -0.402326 | 0.022240         | -0.248724    |
| Inheritance Depth           | -0.434877 | -0.001126        | -0.245719    |
| Internal Function Count     | -0.261861 | 0.004516         | -0.306969    |
| Max Local Variables         | -0.437025 | 0.033653         | -0.365867    |
| Mean Cyclomatic Complexity  | -0.187950 | 0.030606         | -0.285589    |
| Mean Local Variables        | -0.199902 | 0.023715         | -0.288506    |
| Number of Functions         | -0.547065 | 0.015417         | -0.473638    |
| Private Function Count      | -0.290295 | 0.008278         | -0.176321    |
| Public Function Count       | -0.371276 | 0.022805         | -0.206357    |
| State Variable Count        | -0.451398 | 0.033631         | -0.114680    |
| Total Cyclomatic Complexity | -0.295900 | 0.026791         | -0.317429    |

|                             | Stoppable | Storage Saver | Supporter |
|-----------------------------|-----------|---------------|-----------|
| Contract Lines of Code      | 0.074723  | -0.195560     | -0.049936 |
| External Function Count     | -0.027970 | -0.172582     | -0.588692 |
| Inheritance Depth           | -0.000315 | -0.274832     | -0.422558 |
| Internal Function Count     | 0.001070  | -0.207488     | 0.027824  |
| Max Local Variables         | -0.023514 | -0.202206     | -0.333875 |
| Mean Cyclomatic Complexity  | 0.095975  | -0.154593     | 0.117241  |
| Mean Local Variables        | -0.055043 | -0.033436     | -0.189651 |
| Number of Functions         | -0.002143 | -0.245871     | -0.356748 |
| Private Function Count      | -0.000353 | -0.181778     | -0.308781 |
| Public Function Count       | 0.039205  | -0.300203     | -0.375274 |
| State Variable Count        | 0.152770  | -0.337146     | -0.180477 |
| Total Cyclomatic Complexity | 0.047470  | -0.262045     | -0.065596 |



### 1.4.3 Kruskal-Wallis Test

```
[35]: def is_normal(data):
    _, p = stats.normaltest(data, nan_policy='omit')
    return p > 0.05 # p > 0.05 means data is approximately normal

def kruskal_wallis():
    significant_metrics = []
    for metric in metrics:
        groups = [group[metric] for name, group in combined_metrics_df.
        ↳groupby('Blockchain')]
        h_stat, p_val = stats.kruskal(*groups)
        print(f"{metric}: Kruskal-Wallis H = {h_stat:.2f}, p = {p_val:.4e}
        ↳(Non-parametric)")
        if p_val < 0.05:
            significant_metrics.append(metric)
    return significant_metrics

kw_metrics = kruskal_wallis()
```

Contract Lines of Code: Kruskal-Wallis H = 1517.30, p = 0.0000e+00 (Non-parametric)

External Function Count: Kruskal-Wallis H = 4188.78, p = 0.0000e+00 (Non-

```

parametric)
Inheritance Depth: Kruskal-Wallis H = 8233.39, p = 0.0000e+00 (Non-parametric)
Internal Function Count: Kruskal-Wallis H = 1011.38, p = 1.2219e-217 (Non-
parametric)
Max Local Variables: Kruskal-Wallis H = 4437.89, p = 0.0000e+00 (Non-parametric)
Mean Cyclomatic Complexity: Kruskal-Wallis H = 1273.76, p = 1.6278e-274 (Non-
parametric)
Mean Local Variables: Kruskal-Wallis H = 1716.99, p = 0.0000e+00 (Non-
parametric)
Number of Functions: Kruskal-Wallis H = 1608.55, p = 0.0000e+00 (Non-parametric)
Private Function Count: Kruskal-Wallis H = 2900.88, p = 0.0000e+00 (Non-
parametric)
Public Function Count: Kruskal-Wallis H = 3703.15, p = 0.0000e+00 (Non-
parametric)
State Variable Count: Kruskal-Wallis H = 3534.40, p = 0.0000e+00 (Non-
parametric)
Total Cyclomatic Complexity: Kruskal-Wallis H = 710.17, p = 2.1895e-152 (Non-
parametric)

```

#### 1.4.4 Dunn's Test

```

[36]: def dunns_test():
    dunn_results = {}
    for metric in kw_metrics:
        posthoc_matrix = posthoc_dunn(combined_metrics_df, val_col=metric,
        ↪group_col='Blockchain', p_adjust='bonferroni')
        dunn_results[metric] = posthoc_matrix

    # Display Dunn's test results
    for metric, result in dunn_results.items():
        plt.figure(figsize=(8, 6))
        sns.heatmap(result, annot=True, fmt=".2f", cmap="coolwarm", vmin=0,
        ↪vmax=1, linewidths=0.5,
                        cbar_kws={'label': 'p-value'})

        plt.title(f"Dunn's Test for {metric}")
        plt.xlabel("Blockchain")
        plt.ylabel("Blockchain")
        plt.xticks(rotation=45, ha="right")
        plt.yticks(rotation=0)

    plt.show()

dunns_test()

```

